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SYSTEM AND METHODS FOR ELECTRIC VEHICLE CHARGING VIA A PUSH REMOTE PAYMENT TRANSACTION

Abstract

Techniques for enabling secure remote commerce (SRC) transactions for electric vehicle (EV) charging at a charge point (CP) are disclosed. The techniques include establishing a mutually authenticated and encrypted communication session between an EV user system and the CP using transport layer security (TLS). The EV receives merchant information, including a charge point identifier and charge point operator certificate details. The EV retrieves payment credentials from a digital wallet, including a digital card ID assigned by an SRC system. A cryptographically signed payload containing the digital card ID, merchant information, and a timestamp is generated and signed using a private key associated with the EV. The signed payload is transmitted securely to the SRC system via an application programming interface (API). Upon receiving an authorization response approving the payment transaction, the EV transmits a command to the CP to initiate charging.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS [0001] The present application claims priority from U.S. Provisional Patent Application No. 63/556,777, filed Feb. 22, 2024, and titled SYSTEM AND METHODS FOR ELECTRIC VEHICLE CHARGING; U.S. Provisional Patent Application No. 63/557,143, filed Feb. 23, 2024, and titled SYSTEM AND METHODS FOR ELECTRIC VEHICLE CHARGING; and U.S. Provisional Patent Application No. 63/719,863, filed Nov. 13, 2024, and titled SYSTEM AND METHODS FOR ELECTRIC VEHICLE CHARGING. The entire disclosure, including any appendices, of each of the aforementioned priority applications is hereby incorporated by reference herein.

FIELD OF THE DISCLOSURE

[0002] The field of the disclosure relates generally to systems and methods for charging an electric vehicle and, more particularly, to systems and methods for obtaining transaction information directly from the electric vehicle.

BACKGROUND OF THE DISCLOSURE

[0003] In recent years, the number of disparate electric vehicle charging stations has increased. Each of the different charging stations may require the use of different applications in order to authorize charging of electric vehicles and exchange transaction information.

[0004] Recent standards, such as ISO 15118, define a vehicle-to-grid (V2G) communication interface for bi-directional charging and discharging of electric vehicles. Additionally, open loop payments may be provided over the V2G communication interface. When performing open loop payments, transaction information may be exchanged between an electric vehicle (EV) and a charging point (CP) over the same wire used to charge the EV. After the transaction information is exchanged, the CP provides a charging current to the EV. Currently, charge point operators (CPOs) use varying, often incompatible, payment systems, preventing consumers from charging at certain charge points (CPs). Further, when the EV attempts to connect to a CP that is not compatible with a payment system currently used by the customer, the customer may be required to perform cumbersome steps required to register for a new payment system used by the CP and provide payment information to the new CP. Furthermore, data privacy concerns may arise when a customer is required to provide their payment information to a number of different CPOs.

BRIEF DESCRIPTION OF THE DISCLOSURE

[0005] This brief description is provided to introduce a selection of concepts in a simplified form that are further described in the detailed description below. This brief description is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to be used to limit the scope of the claimed subject matter. Other aspects and advantages of the present

disclosure will be apparent from the following detailed description of the embodiments and the accompanying figures.

[0006] In one aspect, a computer-implemented method for enabling secure remote commerce (SRC) transactions for electric vehicle (EV) charging at a charge point (CP) is provided. The method includes establishing, by a user system of the EV, a mutually authenticated and encrypted communication session with the CP via a transport layer security (TLS) protocol. The method further includes receiving, by the user system, merchant information from the CP, wherein the merchant information includes a unique identifier (ID) of the CP and a charge point operator certificate issuer identifier. The method also includes retrieving, by the user system, payment credentials from a digital wallet stored in the EV, the payment credentials comprising a digital card ID assigned to the EV by an SRC system. Additionally, the method includes generating, by the user system, a cryptographically signed payload comprising the digital card ID, the merchant information, and a timestamp, wherein the signed payload is signed by the user system using a private key associated with the EV. The method includes transmitting, by the user system, the signed payload to the SRC system via an application programming interface (API) over a secure communication channel. The method further includes receiving, by the user system, an authorization response from the CP, wherein the authorization response indicates that the payment transaction is approved. In response to receiving the authorization approval, the method includes transmitting, by the user system, a command to the CP to initiate charging of the EV.

[0007] In another aspect, a computing system for enabling a secure remote commerce (SRC) transaction for charging an electric vehicle (EV) at a charging point (CP) is provided. The system includes one or more processors and a memory having computer-executable instructions stored thereon. When executed by the one or more processors, the instructions cause the processors to perform operations to facilitate the secure transaction. The one or more processors establish a mutually authenticated and encrypted communication session with the CP via a transport layer security (TLS) protocol. The one or more processors further receive merchant information from the CP, wherein the merchant information includes a unique identifier (ID) of the CP and a charge point operator certificate issuer identifier. The one or more processors retrieve payment credentials from a digital wallet stored in the EV, the payment credentials comprising a digital card ID assigned to the EV by an SRC system. The one or more processors also generate a cryptographically signed payload comprising the digital card ID, the merchant information, and a timestamp, wherein the signed payload is signed by the user system using a private key associated with the EV. The one or more processors transmit the signed payload to the SRC system via an application programming interface (API) over a secure communication channel. The one or more processors receive an authorization response from the CP, wherein the authorization response indicates that the payment transaction is approved. In response to receiving the authorization approval, the one or more processors transmit a command to the CP to initiate charging of the EV.

[0008] A variety of additional aspects will be set forth in the detailed description that follows. These aspects can relate to individual features and to combinations of features. The advantages of these and other aspects will be apparent to those skilled in the art from the following description of the exemplary embodiments which have been shown and described by way of illustration. As will be realized, the details of the present aspects described herein may be modified in various respects. Accordingly, the figures and description are to be regarded as illustrative in nature and not as restrictive.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] The figures described below depict various aspects of systems and methods disclosed

therein. It should be understood that each figure depicts an embodiment of a particular aspect of the disclosed systems and methods, and that each of the figures is intended to accord with a possible embodiment thereof. Further, wherever possible, the following description refers to the reference numerals included in the following figures, in which features depicted in multiple figures are designated with consistent reference numerals.

[0010] FIG. 1 is a block diagram of an example vehicle charging payment network system;

[0011] FIG. 2 is an example configuration of an electric vehicle, such as the electric vehicle of the vehicle charging payment network system shown in FIG. 1;

[0012] FIG. 3 is an example configuration of a charge point operated by a charge point operator, such as the merchant of the vehicle charging payment network system shown in FIG. 1;

[0013] FIG. 4 is an example configuration of a user system, such as the user system shown in FIG. 1;

[0014] FIG. 5 is an example configuration of a server system, such as the server system shown in FIG. 1;

[0015] FIG. 6 is a process flow diagram for one embodiment of registering an electric vehicle with the vehicle charging payment network system shown in FIG. 1;

[0016] New FIG. 7 is a sequence diagram of a method for generating a contract and provisioning the contract to the electric vehicle of the vehicle charging payment network system shown in FIG. 1;

[0017] FIG. 8 is a process flow diagram for binding payment card information with a digital wallet of an electric vehicle utilizing the vehicle charging payment network system shown in FIG. 1;

[0018] FIG. 9 is a schematic diagram for one embodiment of the vehicle charging payment network system shown in FIG. 1, showing data flow among the parties during a payment transaction for charging an electric vehicle;

[0019] FIG. 10 is a schematic diagram for an alternative embodiment of the vehicle charging payment network system shown in FIG. 1, showing data flow among the parties during a payment transaction for charging an electric vehicle;

[0020] FIG. 11 is a schematic diagram for another alternative embodiment of the vehicle charging payment network system shown in FIG. 1, showing data flow among the parties during a payment transaction for charging an electric vehicle;

[0021] FIG. 12 is a process flow diagram of a process for transferring data and electrical power between an electric vehicle and a charge point;

[0022] FIGS. 13A and 13B are process flow diagrams of an alternative process for transferring data and electrical power between an electric vehicle and a charge point;

[0023] FIGS. 14A and 14B are process flow diagrams of another alternative process for transferring data and electrical power between an electric vehicle and a charge point;

[0024] FIGS. 15A and 15B are process flow diagrams of another alternative process for transferring data and electrical power between an electric vehicle and a charge point; and

[0025] FIG. 16 is a schematic diagram for another alternative embodiment of the vehicle charging payment network system shown in FIG. 1, showing data flow among the parties during a payment transaction for charging an electric vehicle.

[0026] Unless otherwise indicated, the figures provided herein are meant to illustrate features of embodiments of this disclosure. These features are believed to be applicable in a wide variety of systems comprising one or more embodiments of this disclosure. As such, the figures are not meant to include all conventional features known by those of ordinary skill in the art to be required for the practice of the embodiments disclosed herein.

DETAILED DESCRIPTION OF THE DISCLOSURE

[0027] The following detailed description of embodiments of the invention references the accompanying figures. The embodiments are intended to describe aspects of the invention in sufficient detail to enable those with ordinary skill in the art to practice the invention. The

embodiments of the invention are illustrated by way of example and not by way of limitation. Other embodiments may be utilized and changes may be made without departing from the scope of the claims. The following description is, therefore, not limiting. It is contemplated that the invention has general application to charging electric vehicles. The scope of the present invention is defined only by the appended claims, along with the full scope of equivalents to which such claims are entitled.

Example Electric Vehicle Charging Payment System

[0028] FIG. 1 is a block diagram of an example electric vehicle (EV) charging payment network system **10**. In the exemplary embodiment, the system **10** facilitates providing interchange network services offered by an interchange network **16** (broadly, a “payment processor”). In addition, the system **10** enables payment card transactions in which merchants **12**, acquirers **14**, issuers **18**, and/or e-mobility service providers (eMSPs) **28** do not need to have a one-to-one relationship. Although parts of the system **10** are presented in one arrangement, other embodiments may include the same or different parts arranged otherwise, depending, for example, on authorization processes for purchase transactions, communication between computing devices, etc.

[0029] As used herein, the phrase “payment card network” or “interchange network” includes a system or network used for the transfer of funds between two or more parties using cash-substitutes. Transactions performed via a payment card network may include, for example, goods and/or service purchases, credit purchases, debit transactions, fund transfers, account withdrawals, and the like. Payment card networks may be configured to perform such transactions using cash-substitutes including, for example, and without limitation, payment cards, checks, financial accounts, and the like. The phrases payment card network and/or interchange network may include the payment card network as an entity, and the physical payment card network, such as the equipment, hardware, and software making up the network.

[0030] In the example embodiment, the system **10** generally includes the merchant **12**, the acquirer **14**, the interchange network **16**, the issuer **18**, and the eMSP **28** coupled together in communication via a network **20**. The network **20** includes, for example and without limitation, one or more of a local area network (LAN), a wide area network (WAN) (e.g., the Internet, etc.), a mobile network, a virtual network, and/or any other suitable public and/or private network capable of facilitating communication among the merchant **12**, the acquirer **14**, the interchange network **16**, the eMSP **28**, and/or the issuer **18**. In some embodiments, the network **20** may include more than one type of network, such as a private payment transaction network provided by the interchange network **16** to the acquirer **14**, the issuer **18** and the eMSP **28**, and, separately, the public Internet, which may facilitate communication between the merchant **12**, the interchange network **16**, the acquirer **14**, the eMSP **28**, and consumers **22**, etc.

[0031] Embodiments described herein may relate to a transaction card system, such as a credit card payment system using the Mastercard® interchange network. (Mastercard is a registered trademark of Mastercard International Incorporated.) The Mastercard interchange network is a set of proprietary communications standards promulgated by Mastercard International Incorporated for the exchange of financial transaction data and the settlement of funds between financial institutions that are members of the Mastercard interchange network. As used herein, “financial transaction data” includes both (1) a unique account number—such as a primary account number (PAN)—that is linked to an account holder or consumer **22** using a payment card issued by an issuer, and (2) purchase data reflecting the consumer's transaction. This purchase data may include details such as the merchant type, purchase amount, purchase date, and additional relevant information. Such data may be transmitted among any parties within the network system **10**.

[0032] In a typical transaction card system, a financial institution (the “issuer”) issues a transaction card, such as a credit card, to a consumer, such as the consumer **22**. The consumer **22** uses the transaction card to tender payment for a purchase from the merchant **12**. The consumer **22** may input information from a transaction card (broadly, payment credentials) into a user system **32** and

store the information in a digital wallet **40** as digital wallet data. The merchant **12** is typically associated with products, for example, and without limitation, goods and/or services, that are offered for sale and are sold to the consumer **22**. In particular, the merchant **12** is associated with operating charge points (CPs) **34**. The CPs **34** are configured to provide electrical energy to electric vehicles (EVs). The merchant **12** includes, for example, a physical location and/or a virtual location such as an Internet-based storefront.

[0033] To accept payment from the consumer **22** using, for example, the digital wallet data stored in the digital wallet **40**, the merchant **12** must normally establish an account with a financial institution that is part of the system **10**. This financial institution is usually called the “merchant bank,” the “acquiring bank,” or the acquirer **14**. When the consumer **22** submits payment for a purchase with the user system **32** using the digital wallet **40**, the merchant **12** requests authorization from the acquirer **14** for the purchase. The request may be performed over a telephone but is usually performed using a point-of-sale terminal that reads the consumer's account information from a magnetic stripe, a chip, embossed characters on the transaction card, or digital wallet data and communicates electronically with the transaction processing computers of the acquirer **14**. Alternatively, the acquirer **14** may authorize a third party to perform transaction processing on its behalf. In this case, the point-of-sale terminal will be configured to communicate with the third party. Such a third party is usually called a “merchant processor,” an “acquiring processor,” or a “third party processor.”

[0034] The interchange network **16** and computers of the acquirer **14** or merchant processors will communicate with computers of the issuer **18** to determine whether the consumer's account is in good standing and whether the purchase is covered by the consumer's available credit line. Based on these determinations, the request for authorization will be declined or accepted. If the request is accepted, an authorization code is issued to the merchant **12**.

[0035] When a request for authorization is accepted, the available credit line of the consumer's account is decreased. Normally, a charge for a payment card transaction is not posted immediately to the consumer's account because bankcard associations, such as Mastercard, have promulgated rules that do not allow the merchant **12** to charge, or “capture,” a transaction until the purchased goods are shipped or the purchased services are delivered. However, with respect to at least some debit card transactions, a charge may be posted at the time of the transaction. When the merchant **12** ships or delivers the goods or services, the merchant **12** captures the transaction by, for example, appropriate data entry procedures on the point-of-sale terminal. This may include bundling of approved transactions daily for standard retail purchases. If the consumer **22** cancels a transaction before it is captured, a “void” is generated. If the consumer **22** returns goods after the transaction has been captured, a “credit” is generated. The interchange network **16** and/or the issuer **18** stores in a database, such as database **26**, the payment card information, such as, and without limitation, a type of merchant, a merchant identifier, a location where the transaction was completed, an amount of purchase, and a date and time of the transaction.

[0036] After a purchase has been made, a clearing process occurs to transfer additional transaction data related to the purchase among the parties to the transaction, such as the acquirer **14**, the interchange network **16**, and the issuer **18**. More specifically, during and/or after the clearing process, additional data, such as a time of purchase, a merchant name, a type of merchant, purchase information, consumer account information, a type of transaction, itinerary information, information regarding the purchased item and/or service, and/or other suitable information, is associated with a transaction and transmitted between parties to the transaction as transaction data, and may be stored by any of the parties to the transaction.

[0037] After a transaction is authorized and cleared, the transaction is settled among the merchant **12**, the acquirer **14**, and the issuer **18**. Settlement refers to the transfer of financial data or funds among the merchant **12**, the acquirer **14**, and the issuer **18** related to the transaction. Usually, transactions are captured and accumulated into a “batch,” which is settled as a group. More

specifically, a transaction is typically settled between the issuer **18** and the interchange network **16**, and then between the interchange network **16** and the acquirer **14**, and then between the acquirer **14** and the merchant **12**.

[0038] In some embodiments, the payment card transaction is a card present transaction conducted, for example, by swiping or dipping a payment card at the merchant's point-of-sale (POS) terminal. Alternatively, the payment card transaction may be a card-not-present transaction conducted, for example, with a payment card stored on file with the merchant or stored as digital wallet data in a digital wallet **40** on a consumer's computing device, such as the user system **32**. In some embodiments, the user system **32** and digital wallet **40** may be integrated with the electric vehicle (EV) **24**.

[0039] The user system **32** may be, for example, a computing device integrated with the EV **24**, a cellular telephone, a smart watch or other electronic wearable apparel, a tablet, an implanted smart device, a personal computing device, or any other electronic device capable of two-way digital communications which may be associated with a consumer. In some embodiments, the user system **32** may be replaced with another computing device suitable for performing the functions disclosed herein (e.g., a desktop or laptop computer, a smart television, etc.).

[0040] In the exemplary embodiment, the user system **32** may also allow the consumer **22** to create an account with a server system **30** or with another component of the interchange network **16**.

During the account setup process, the consumer **22** may send account registration information to the server system **30** via the user system **32**. The account registration information may include, for example, and without limitation, payment account data (e.g., the PAN, a virtual payment number, limited use number, etc.) and user system identification data (e.g., a provisioning certificate identifier (PCID), Electronic Serial Number (ESN), Mobile Equipment Identifier (MEID), International Mobile Equipment Identity (IMEI) number, and the like).

[0041] For instance, the server system **30** may receive account registration information from the user system **32** (identifying the user system **32**) and a payment account or PAN associated with the consumer **22**. The consumer **22** may, for example, set up the account with the server system **30** by providing the account registration information and generating a login identifier (i.e., a UserID) and a password used when logging into an application for communicating with the server system **30**. The consumer **22** may transmit various information or data to the server system **30** via the application, which may be stored on, partially stored on, or accessed via a web-browser of the user device **32**. The information or data transmitted by the user to the server system **30** may include, for example, and without limitation, authentication information associated with the consumer's PAN, biometrics of the consumer, and/or contact information. The contact information may include one or more ways to communicate with the consumer **22**, including, for example, a push notification associated with the mobile application, a short messaging service (SMS) message, an email message, telephone number, and the like. The server system **30** may generate a new account profile or update an existing account profile for the account associated with the account registration information received from the user system **32**. Payment information associated with the account is stored in the digital wallet **40**.

[0042] In the example embodiment, the system **10** includes the server system **30**, which may be part of the interchange network **16** (shown in FIG. 1). The server system **30** is coupled in communication with the CPs **34**, the merchant **12**, the acquirer **14**, and the issuer **18**. The server system **30** is also coupled in communication with a one or more client systems, also referred to as the user systems **32**. In one embodiment, the user systems **32** are computers including a web browser, such that server system **30** is accessible to the user systems **32** using the Internet. The user systems **32** are interconnected to the Internet through one or more of many interfaces including, for example, a network, such as a LAN or WAN, dial-in-connections, cable modems, and/or special high-speed Integrated Services Digital Network (ISDN) lines. The user systems **32** could be any device capable of interconnecting to the Internet including an Internet connected phone, a PDA, or

any other suitable web-based connectable equipment.

[0043] In the example embodiment, the system **10** includes the one or more CPs **34**, which may be connected to the merchant **12** and to the server system **30**. The CPs **34** may be interconnected to the Internet (or any other network that allows the CPs **34** to communicate as described herein) through many interfaces including a network, such as a local area network (LAN) or a wide area network (WAN), dial-in-connections, cable modems, wireless modems, and special high-speed ISDN lines. The CPs **34** are any device capable of interconnecting to the Internet and including an input device capable of reading information received from the digital wallet **40** of the user system **32**. As used herein, the term charge point is used broadly and generally to refer to any device capable of charging an electric vehicle in which a consumer interacts with a merchant to complete a payment card transaction.

[0044] In the example embodiment, the server system **30** is connected to a database **26**. The database **26** is configured to store information on a variety of matters, including account information associated with consumers. In one embodiment, the database **26** is a centralized database stored on the server system **30** and can be accessed by potential users at the user system **32** by logging onto the server system **30** through the user system **32**. In an alternative embodiment, the database **26** is stored remotely from the server system **30** and may be a distributed or non-centralized database.

[0045] In one example embodiment, the database **26** may include a single database having separated sections or partitions or may include multiple databases, each being separate from each other. The database **26** may store transaction data generated as part of sales activities and savings activities conducted over the processing network including data relating to merchants, account holders or customers, issuers, acquirers, savings amounts, savings account information, and/or purchases made. The database **26** may also store account data including at least one of a consumer name, a consumer address, an account number, and other account identifiers. The database **26** may also store merchant data including a merchant identifier that identifies each merchant registered to use the network, and instructions for settling transactions including merchant bank account information. The database **26** may also store purchase data associated with items being purchased by a consumer from a merchant, and authorization request data. The database **26** may also store device information, payment card information, and other data involved with processing transactions between one or more parties.

[0046] In the example embodiment, the user system **32** may be associated with the consumer **22** and the EV **24**. Alternatively, the user system **32** may be associated with the merchant **12**, issuer **18**, and/or the acquirer **14**. The CP **34** may be associated with the merchant **12** (shown in FIG. **1**). The server system **30** may be associated with the interchange network **16** or a payment processor. In the example embodiment, the server system **30** is associated with a financial transaction processing network, such as the interchange network **16**, and may be referred to as an interchange computer system. The server system **30** may be used for processing transaction data. In addition, the user system **32** and the CP **34** may include a computer system associated with at least one of a merchant, an online bank, a bill payment outsourcer, an acquirer bank, an acquirer processor, an issuer bank associated with a payment card, an issuer processor, a remote payment processing system, and/or a biller. It is noted that the system **10** may include more, fewer, or alternative components and/or perform more, fewer, or alternative actions, including those discussed elsewhere herein.

[0047] In the exemplary embodiment, the eMSP **28** facilitates enabling the consumer **22** to access multiple CPs **34** (operated by different merchants **12**, for example) without the need to have a membership or account with the specific CP **34** or merchant **12**. For example, the eMSP **28** may streamline one or more services, such as billing, customer support, and network management. In some embodiments, the eMSP **28** may function as an “aggregator platform” for electric vehicle charging. That is, the eMSP **28** may facilitate interaction between EV drivers (i.e., the consumer **22**) and various charging networks (i.e., CPs **34** and merchants **12**), eliminating the need for

separate memberships at each CP **34**. As described further herein, the eMSP **28** provides payment contract services for the consumer **22** to enable secure plug-and-charge functionality. The eMSP **28** includes a computing system **36**. The computing system **36** includes, for example, a provisioning certificate pool (PCP) module **42**, an enrollment over secure transport (EST) for V2G module **44**, a certificate provisioning service (CPS) module **46**, and a contract certificate pool (CCP) module **48**. These modules are collectively responsible for managing, signing, and distributing cryptographic credentials and contract data to enable secure, automated payment-based charging for electric vehicles (EVs), as described herein.

Exemplary Computer Systems

[0048] FIG. **2** is an example configuration of an electric vehicle (EV) **200**, such as the EV **24** (shown in FIG. **1**). In the example embodiment, the EV **200** includes an integrated user system **32**, an electrical connector **220**, and a battery **230**. In some embodiments, the user system **32** includes the digital wallet **40**. In the example embodiment, the electrical connector **220** may be integrated with the battery **230** of the EV **200** or may be connected to a charging cable **210** attached to the battery **230** of the EV **200**.

[0049] In the example embodiment, the electrical connector **220** is configured to be connected to a corresponding electric connector of a charge point, such as an electric connector **320** (shown in FIG. **3**). Once the electrical connector **220** is connected to the corresponding connector of the charge point, electronic communication may be established between the user system **32** of the EV **200** and a computing device of the charge point. In the example embodiment, both data and electrical power may be transferred over the charging cable **210** and the connector **220**.

[0050] In some embodiments, the charge point and/or the EV **200** communicates with various components of the system **10** (shown in FIG. **1**) to authorize a transaction enabling a transfer of electrical energy from the charge point to the EV **200**. In some embodiments, the EV **200** may be configured to receive, for example, one hundred and twenty (120) volts of alternating current (VAC), two hundred and forty (240) VAC, direct current (DC) power, or any other form of electrical energy from a corresponding charge point.

[0051] Electrical energy that is received by the EV **200** is provided to the battery **230** as a charging current. In some embodiments, the EV **200** may include various circuitry components (not shown), such as an input filter, a rectifier, one or more DC/DC converters, current sensors, voltage sensors, fault protection circuitry, and/or any other circuitry necessary to provide the charging current to the battery **230**.

[0052] In some embodiments, the electrical connector **220** may be an electrical plug configured to be received by a corresponding receptacle of a charge point, such as the CP **34** (shown in FIG. **1**). Alternatively, the connector **220** may be a receptacle configured to receive a corresponding electrical plug of the charge point. In an alternative embodiment, the connector **220** may be an inductive charging coil configured to wirelessly transfer data and electrical energy.

[0053] In some embodiments, the battery **230** may include one or more lithium-ion batteries, nickel-metal hydride batteries, lead-acid batteries, ultracapacitors a capacitor bank, and the like. The battery **230** may provide power to various electronic components of the EV **200** including at least one or more electric motors (not shown), a powertrain system (not shown), one or more auxiliary batteries (not shown), an electronic control unit (not shown), a vehicle display (not shown), the user system **32**, and any other electronic circuitry requiring electric power as part of the operation of the EV **200**.

[0054] In the example embodiment, the user system **32** is integrated with the EV **200**. The user system **32** includes the digital wallet **40**. The digital wallet **40** is configured to store payment account information for one or more payment accounts of the consumer **22**. In some embodiments, upon detecting a connection with a charge point, the user system **32** establishes communication with the connected charge point. The charge point sends merchant information to the EV **200**. Subsequently, the EV **200** retrieves payment information from the digital wallet **40** and provides

the payment information and the merchant information to the connected charge point via a communication interface, such as a communication interface **410** (shown in FIG. **4**). After the payment information and the merchant information are processed, the EV **200** begins to receive a charging current from the charge point.

[0055] FIG. **3** is an example configuration of a charge point (CP) **300** operated by a charge point operator (CPO), such as the merchant **12** of the vehicle charging payment network system **10** shown in FIG. **1**. In some embodiments, the CP **300** is one of the CPs **34** (shown in FIG. **1**). In the example embodiment, the CP **300** includes a charging cable **310**, an electrical connector **320**, and a computing system **330**. The charging cable **310** and electrical connector **320** are configured to convey both data and electrical energy between the CP **300** and an electric vehicle, such as the EV **200** (shown in FIG. **2**). The electrical connector **320** is configured to connect with a corresponding electrical connector of an EV, such as the connector **220** of the EV **200** (shown in FIG. **2**). The computing system **330** is configured to process data and communicate with various components of the system **10** (shown in FIG. **1**).

[0056] In some embodiments, the electrical connector **320** may be a receptacle configured to receive a corresponding electrical connector of an electric vehicle, such as the electrical connector **220** of the EV **200** (shown in FIG. **2**). In other embodiments, the electrical connector **320** may be a plug configured to be received by a corresponding receptacle of an EV, such as the electrical connector **220** of the EV **200** (shown in FIG. **2**). In an alternative embodiment (not shown), the electrical connector **320** may be integrated in the CP **300** without the use of a charging cable. In another alternative embodiment, the electrical connector **320** may be an inductive charging coil configured to wirelessly transfer data and electrical energy. In some embodiments, the CP **300** may be configured to provide one hundred and twenty (120) volts of alternating current (VAC), two hundred and forty (240) VAC, direct current (DC), or any other form of electrical energy to a corresponding EV.

[0057] In the example embodiment, the computing system **330** includes a communication interface, such as the communication interface **410** (shown in FIG. **4**), to communicate with an EV, such as the EV **200** (shown in FIG. **2**) and an associated CPO, such as the merchant **12** (shown in FIG. **1**). In some embodiments, the computing system **330** provides merchant information, including unique identifiers associated with the CP and the CPO, to the EV. The EV returns account information and the merchant information to the CP. The CP then forwards the transaction information to the CPO. The CPO authorizes the transaction and instructs the CP to begin charging the EV. Subsequently, the CP begins providing charging current to the EV. The communication interface may include, for example, a wired or wireless network adapter or a wireless data transceiver for use with Bluetooth communication, radio frequency communication, near field communication (NFC), and/or with a mobile phone network, Global System for Mobile communications (GSM), 3G, or other mobile data network, and/or Worldwide Interoperability for Microwave Access (WiMax) and the like.

[0058] FIG. **4** is an example configuration of a computing system **400**. In some embodiments, the computing system **400** is a user system **32** (shown in FIG. **1**) operated by a user **401**, such as the consumer **22** (shown in FIG. **1**). The computing system **400** is also or alternately the CP computing system **330** (shown in FIG. **3**). In some embodiments, the computing system **400** is integrated with the EV **24** (shown in FIG. **1**). In the example embodiment, the computing system **400** includes a processor **402** for executing instructions. In some embodiments, executable instructions are stored in a memory device **404**. The processor **402** includes one or more processing units in, for example, a multi-core configuration. The memory device **404** is any device allowing information such as digital wallet data, executable instructions, and/or written works to be stored and retrieved. The memory device **404** includes one or more computer readable media.

[0059] A location of the computing system **400** can be obtained through conventional methods, such as a location service (e.g., global positioning system (GPS) service) in the computing system **400**, “ping” data that includes geotemporal data, from cell location register information held by a

telecommunications provider to which the computing system **400** is connected, and the like. For example, in one suitable embodiment, a GPS chip can be part of or separate from the processor **302** to enable the location of the computing system **400** to be determined.

[0060] The computing system **400** also includes at least one media output component **406** for presenting information to the user **401**. The media output component **406** is any component capable of conveying information to the user **401**. In some embodiments, the media output component **406** includes an output adapter such as a video adapter and/or an audio adapter. An output adapter is operatively coupled to the processor **402** and operatively connectable to an output device such as a display device, a liquid crystal display (LCD), organic light emitting diode (OLED) display, or “electronic ink” display, or an audio output device, a speaker, or headphones.

[0061] In some embodiments, the computing system **400** includes an input device **408** for receiving input from the user **401**. The input device **408** may include, for example, a touch sensitive panel, a touch pad, a touch screen, a stylus, a gyroscope, an accelerometer, a position detector, a keyboard, a pointing device, a mouse, or an audio input device. A single component such as a touch screen may function as both an output device of the media output component **406** and the input device **408**. The computing system **400** may also include a communication interface **410**, which is communicatively connectable to a remote device such as the server system **30** (shown in FIG. 1), the EV **24** (shown in FIG. 1) and/or the CP **34** (shown in FIG. 1). The communication interface **410** may include, for example, a wired or wireless network adapter or a wireless data transceiver for use with Bluetooth communication, radio frequency communication, near field communication (NFC), and/or with a mobile phone network, Global System for Mobile communications (GSM), 3G, or other mobile data network, and/or Worldwide Interoperability for Microwave Access (WiMax) and the like.

[0062] Stored in the memory device **404** are, for example, computer readable instructions for providing a user interface to the user **401** via the media output component **406** and, optionally, receiving and processing input from the input device **408**. A user interface may include, among other possibilities, a web browser and a client application. Web browsers enable users, such as the user **401**, to display and interact with media and other information typically embedded on a web page or a website from the server system **30**. A client application allows the user **401** to interact with a server application from the server system **30**. In the example embodiment, the memory device **404** may store digital wallet data corresponding to a digital wallet, such as the digital wallet **40** (shown in FIG. 1).

[0063] In the example embodiment, the computing device **400** is a user computing device from which the user **401** engages with a merchant (e.g., the merchant **12** shown in FIG. 1), an interchange network (e.g., the interchange network **16** shown in FIG. 1), and an issuer of a payment card (e.g., the issuer **18** shown in FIG. 1) to perform a payment transaction that undergoes an additional authentication process.

[0064] FIG. 5 is an example configuration of a server system **500**, such as the server system **30** or eMSP computing system **28** (each shown in FIG. 1). In the example embodiment, the server system **500** includes a processor **502** for executing instructions. The instructions may be stored in a memory area **504**, for example. The processor **502** includes one or more processing units (e.g., in a multi-core configuration) for executing the instructions. The instructions may be executed within a variety of different operating systems on the server system **500**, such as UNIX, LINUX, Microsoft Windows®, etc. More specifically, the instructions may cause various data manipulations on data stored in a storage device **510** (e.g., create, read, update, and delete procedures), such as the database **26** (shown in FIG. 1). It should also be appreciated that upon initiation of a computer-based method, various instructions may be executed during initialization. Some operations may be required to perform one or more processes described herein, while other operations may be more general and/or specific to a programming language (e.g., C, C#, C++, Java, or other suitable programming languages, etc.).

[0065] The processor **502** is operatively coupled to a communication interface **506** such that the

server system **500** can communicate with a remote device such as the computing system **400** (shown in FIG. **4**) or another server system. For example, the communication interface **506** may receive communications from the electric vehicle **24** or a user system **32** via the Internet, as illustrated in FIG. **2**.

[0066] The processor **502** is operatively coupled to the storage device **510**. The storage device **510** is any computer-operated hardware suitable for storing and/or retrieving data, such as the database **26** (shown in FIG. **1**). In some embodiments, the storage device **510** is integrated in the server system **500**. In other embodiments, the storage device **510** is external to the server system **500** and is similar to the database **26**. For example, the server system **500** may include one or more hard disk drives as the storage device **510**. In other embodiments, the storage device **510** is external to the server system **500** and may be accessed by a plurality of server systems **500**. For example, the storage device **510** may include multiple storage units such as hard disks or solid-state disks in a redundant array of inexpensive disks (RAID) configuration. The storage device **510** may include a storage area network (SAN) and/or a network attached storage (NAS) system.

[0067] In some embodiments, the processor **502** is operatively coupled to the storage device **510** via a storage interface **508**. The storage interface **508** is any component capable of providing the processor **502** with access to the storage device **510**. The storage interface **508** may include, for example, an Advanced Technology Attachment (ATA) adapter, a Serial ATA (SATA) adapter, a Small Computer System Interface (SCSI) adapter, a RAID controller, a SAN adapter, a network adapter, and/or any component providing the processor **502** with access to the storage device **510**.

[0068] The memory area **504** includes, but is not limited to, random access memory (RAM) such as dynamic RAM (DRAM) or static RAM (SRAM), read-only memory (ROM), erasable programmable read-only memory (EPROM), electrically erasable programmable read-only memory (EEPROM), and non-volatile RAM (NVRAM). The above memory types are exemplary only and are thus not limiting as to the types of memory usable for storage of a computer program.

[0069] In the example embodiment, server system **500** is an authentication processing system in communication with one or more of the acquirer **14**, the issuer **18**, and the merchant **12** during a payment card transaction associated with a user, such as the consumer **22** (shown in FIG. **1**). The server system **500** performs checking for account restrictions for accounts (e.g., PANs) initiating a payment transaction and provides authentication services to a consumer associated with the payment transaction.

Payment Card Enrollment with Secure Remote Commerce (SRC) System

[0070] FIG. **6** is a process flow diagram of a method **600** for registering payment information of a user system (e.g. user system **32** shown in FIG. **1**) associated with an electric vehicle (e.g. the EV **24** shown in FIG. **1**) and used in transactions related to charging the electric vehicle. In one embodiment, the user system **32** may be integrated into the EV **24**. In alternative embodiments, the user system **32** may include a mobile computing device, such as a smartphone, tablet, laptop, and the like. The user system **32** accesses a digital wallet application, such as the digital wallet **40** to register or enroll payment information with a Secure Remote Commerce (SRC) environment or framework (e.g., the server system **30** shown in FIG. **1**) to obtain a digital credential (referred to herein as a “DigitalCardID”). The term “DigitalCardID” is used to denote a secure representation of underlying account data that enables transactions in digital wallet systems without exposing the original PAN.

[0071] At step **605**, a secure application programming interface (API) connection is established between the digital wallet **40** of the user system **32** and the SRC system (e.g. the server system **30** shown in FIG. **1**) via a communications interface (e.g. communication interface **410** shown in FIG. **4**) of the user system **32**. The digital wallet **40** interfaces with the SRC system, for example, via one or more secure APIs. The SRC system is configured to communicate with card issuers, token service providers, and other entities necessary for validating the card information and generating the DigitalCardID.

[0072] After communication is established, at step **610**, the server system **30** receives an enrollment request, which includes payment card information, from the digital wallet **40** based on a user selection. For example, in an embodiment, upon requesting to add a new payment card, the user may be prompted to input relevant account data. The user selection may include entering the payment card information into a graphical user interface (GUI) of the user system **32**. The payment card information may include: (1) a PAN, an expiration date, and a card security code associated with a payment card; and (2) associated user information, such as a name, address, phone number, and/or other user information. Alternatively, in certain implementations, the user may select a payment card already on file with an issuer or a third-party wallet aggregator (such as the EV manufacturer, rental car company, fleet manager, etc.). The digital wallet **40** may collect these data elements and prepare them for secure transmission to the server system **30**.

[0073] The digital wallet **40** typically encrypts or otherwise protects the payment card information prior to transmission. The server system **30** may further employ its own transport security protocols, ensuring that sensitive information is only accessible to authorized components. In some embodiments, the server system **30** may generate a cryptographic nonce or session token, which is used to bind the enrollment session to subsequent steps in the process, mitigating risk of replay attacks or data interception.

[0074] At **612**, after the user's payment card information is received by the server system **30**, an enrollment orchestration service interfaces with the issuer **18** (shown in FIG. **1**) or interchange network **16** (shown in FIG. **1**) to perform card validation. The validation process may include account status checks to confirm the payment card is active and in good standing.

[0075] At step **615**, the server system **30** assigns a payment card identifier (the DigitalCardID). Again, the DigitalCardID is associated with the payment card information and serves as a secure proxy for the PAN. The server system **30** may store the DigitalCardID in a database (e.g. the database **26** shown in FIG. **1**). The DigitalCardID may be stored in the database **26** with the payment card information. The DigitalCardID is cryptographically bound to the user's account and contains sufficient metadata to facilitate digital wallet transactions, while preventing exposure of the true PAN during payment processing. The server system **30** sends the DigitalCardID to the user system **32** via the secure API connection.

[0076] Concurrent with generation of the DigitalCardID, the user may undergo authentication to verify identity. In the example, at step **620**, the server system **30** receives an authentication request, as part of the enrollment request, from the user system **32**. When the user submits the enrollment information (e.g., the payment card information and associated user information) through the digital wallet **40**, the server system **30** analyzes the information and determines whether further identity verification is required. In some embodiments, a set of risk evaluation rules or fraud-detection parameters triggers the need for a card issuer-led identity verification (ID&V) process. Upon detecting such a requirement, at step **625**, the server system **30** transmits an ID&V request to the issuer **18**, for example, using a secure communication channel (e.g., mutually authenticated TLS).

[0077] At step **627**, after receiving the ID&V request, the issuer **18** transmits an ID&V challenge to the user, via the user system **32** or digital wallet **40**, for example. More particularly, in an embodiment, the issuer **18** initiates an authentication session, which may include session identifiers, timeouts, and cryptographic nonces to ensure the integrity of the ID&V challenge process. These session parameters facilitate ensuring that subsequent interactions with the user, via the user system **32**, are bound to the specific enrollment process, thereby mitigating the risk of replay or man-in-the-middle attacks. The ID&V challenge may vary depending on the issuer. For example, the issuer **18** may: (1) transmit a one-time passcode (OTP) to the user system **32**, sent via short message service (SMS), push notification, or email; (2) employ push-based authentication by pushing a secure message to the user system **32**, prompting the user to confirm card enrollment within the issuer's mobile application; or (3) employ knowledge-based questions requiring correct

answers to personal or transactional history questions. It is noted that other ID&V challenge processes may be employed, as known to one skilled in the art. The ID&V challenge may be presented to the user through the digital wallet **40** or via a separate authenticated channel (e.g., the issuer's mobile application installed on the user system **32**). If directed through the digital wallet **40**, an API call from the issuer **18** to the server system **30** may provide instructions, which the server system **30** then relays to the digital wallet **40** for display.

[0078] At step **628**, after receiving the ID&V challenge, the user completes the challenge, for example, by providing the requested authentication input (e.g., entering an OTP). In an example, the digital wallet **40** may collect the user input and securely forward it to the issuer **18** (potentially via the server system **30**). As noted above, the transmission may be protected by end-to-end encryption, based on the session parameters described above, to ensure that the user input is inaccessible to unauthorized entities.

[0079] Upon receipt of the ID&V challenge response from the user, the issuer **18** verifies its correctness against the previously generated challenge values or stored user records. For example, if an OTP is used, the issuer **18** verifies the user-supplied code within a prescribed time window. If the verification is successful, the issuer **18** marks the authentication session as verified, logging the outcome and generating an authentication result token or similar data structure containing the verification status (e.g., verification data).

[0080] At step **630**, if authentication is successful, the server system **30** receives an indication of successful authentication from the issuer **18**. For example, the server system **30** may receive the verification data, which may include success status, timestamp, and any relevant risk scoring, from the issuer **18**. The verification data may be accompanied by additional issuer-provided metadata indicating, for example, whether the user system **32** information is consistent with known device profiles or if further checks are recommended. The server system **30** may process this result and update the enrollment session associated with the DigitalCardID.

[0081] If authentication is not successful, the method **600** returns to step **620** if the user chooses to re-submit authentication information. Otherwise, the method **600** ends upon unsuccessful authentication.

[0082] At step **635**, the server system **30** sends verification data to the user system **32** indicating that the authentication was successful. For example, the server system **30** may compile a verification data package containing, for example, a “verified” flag, the newly generated DigitalCardID, and instructions on default card preferences if so designated by the user or the issuer **18**. The server system **30** may transmit the verification data package to the digital wallet **40** using secure messaging protocols (e.g., JSON or XML over TLS).

[0083] At step **640**, the user system **32** stores the DigitalCardID in the digital wallet **40**. For example, the user system **32** or digital wallet **40** may receive the verification data package, extract the DigitalCardID therefrom, and update its internal record of stored payment credentials. In certain embodiments, the digital wallet **40** may be configured to automatically set the newly verified card as the default method of payment. If a default card assignment is indicated, the digital wallet **40** may modify its local or cloud-based credential store to mark the newly provisioned DigitalCardID as the primary card. Such an action ensures that subsequent transactions default to this credential, streamlining the checkout experience.

Contract Generation

[0084] FIG. 7 is a sequence diagram of a method **700** for generating a contract that is bound to the DigitalCardID (e.g. the DigitalCardID described in FIG. 6) and provisioning the contract to the EV (e.g. the EV **24** shown in FIG. 1). Initially, at step **702**, a manufacturer **38** of the EV **24** generates a provisioning certificate identifier (PCID) for the EV **24**. The PCID is a unique digital code that identifies an electric vehicle (EV). The PCID may be used, for example, during a plug-and-charge process, a feature that allows EVs to automatically charge and pay for charging sessions with little to no additional user action. This PCID may be embedded in a certificate or metadata that

references a cryptographic key pair or a unique identifier. At step **704**, the EV manufacturer **38** transmits a copy of the PCID to the eMSP **28**. More particularly, the PCID is transmitted to the eMSP's provisioning certificate pool (PCP) **42**, which functions as a repository for provisioning-related certificates, keys, and/or reference data. In addition, at step **706**, the EV **24** is provisioned with the PCID. The PCID may be stored in a secure hardware component, such as a microcontroller with built-in secure storage, or a Trusted Platform Module (TPM), of a computing device integrated with the EV **24** (e.g., in some embodiments, the user system **32**). By securing the PCID within the EV **24**, the EV **24** can later authenticate to eMSP services with minimal risk of key compromise.

[0085] At step **708**, a user, such as the consumer **22**, enrolls a payment card with the SRC system (e.g. the server system **30** shown in FIG. **1**), as described above in method **600**. The SRC system is recognized by all participating charge point operators (CPOs) or merchants **12**. As described above, the enrollment process generates the DigitalCardID (or tokenized representation of the user's payment method), which is passed to the digital wallet **40**. The digital wallet **40** registers with the SRC system as a digital card facilitator, enabling the digital wallet **40** to request certificates on behalf of the consumer **22** and to bind those certificates to a valid payment credential (i.e., the DigitalCardID). It is noted that subsequent cryptographic operations and certificate requests by the digital wallet **40** may be authenticated against the SRC system's root certificate authority (CA) hierarchy.

[0086] At step **710**, in order to establish a contract for EV charging services, the digital wallet **40** retrieves the PCID directly from the EV **24**. In an embodiment, the EV **24** may transmit the PCID over any secure link established with the digital wallet **40**, such as, a TLS-encrypted Bluetooth or Wi-Fi direct channel, or an on-board diagnostic (OBD) interface protected by additional security layers.

[0087] At step **712**, after obtaining the PCID from the EV **24**, the digital wallet **40** requests a provisioning certificate from the eMSP **28**. For example, in an embodiment, the digital wallet **40** contacts the eMSP's PCP **42** (via an HTTPS-based API, for example, GET/oem/provCerts?pcid=<PCID>) to obtain the provisioning certificate or its associated certificate chain. The PCP **42** may be responsible for maintaining a database of valid certificates used during the initial provisioning steps. At step **714**, in response to the provisioning certificate request, the PCP **42** transmits the provisioning certificate to the digital wallet **40**.

[0088] At step **716**, the digital wallet **40** generates a certificate signing request (CSR). For example, in the example embodiment, the digital wallet **40** may generate a new public/private key pair (i.e., an asymmetric key pair) for the EV's future contract certificate (broadly, a contract). In one example, the private key may be encrypted using the public key from the provisioning certificate (retrieved from the PCP **42**, as described above). For instance, a symmetric key (e.g., generated by the digital wallet **40**) may be used to encrypt the private key, and that symmetric key may be further encrypted by the public key portion of the provisioning certificate. This ensures that only an authorized entity with the corresponding private key to the provisioning certificate can decrypt the EV's private key. After the key pair is generated, the digital wallet **40** may construct the CSR. The CSR may contain the public key, the DigitalCardID, the PCID, an indication of consumer device consumer verification method (CDCVM) capability, and the full qualified domain name of the SRC system (e.g., <https://www.srcsystem.com>). In an embodiment, the CSR may be formatted in accordance with X.509 standards (e.g., PKCS #10) and may include subject alternative name (SAN) extensions referencing an eMobility account identifier (EMAID).

[0089] At step **718**, to obtain a signed leaf certificate, the digital wallet **40** transmits the CSR to the eMSP **28**. For example, the digital wallet **40** may transmit the CSR to the EST for V2G module **44**. The module **44** may enforce a secure data exchange protocol, such as TLS v1.2 or v1.3, to preserve the confidentiality and integrity of the transmitted CSR. In an example, the EST for V2G module **44** may verify the request, check that the PCID references a valid OEM record in the PCP **42**, and

validate the DigitalCardID against the SRC system's Root CA if necessary.

[0090] At step **720**, the eMSP **28** returns a contract certificate and certificate chain to the digital wallet **40**. For example, in an embodiment, upon successful validation, a subordinate CA (CA Sub 1)—which is signed by the SRC System Root CA—issues and signs the EV contract certificate, forming a certificate chain (Leaf.fwdarw.CA Sub 1.fwdarw.SRC System Root CA). The initial “contract” data, including the newly signed leaf certificate and any intermediate certificates, may be returned to the digital wallet **40**. In some implementations, the eMSP **28** may embed or link additional contract terms in a structured format compatible with ISO 15118 (e.g., “Contract Certificates” or “CertificateInstallationReq/Res” messages) or Open Charge Point Protocol (OCPP).

[0091] At step **722**, the digital wallet **40** performs a contract binding process (described in more detail below with respect to method **800**), in which the digital wallet **40** transmits the certificate chain and associated parameters to the bind API of the SRC system. The API call may include the DigitalCardID, the signed certificate, and any additional metadata necessary for contract validation. The SRC system, acting as the trust anchor for identity and payment, validates the chain against its Root CA certificate. After approval, the SRC system may record the association (i.e., that a particular DigitalCardID is authorized to use this contract certificate) in its internal database and returns a confirmation to the digital wallet **40**.

[0092] At step **724**, the digital wallet **40** transmits the bound contract to the eMSP **28**. For example, in an embodiment, the CPS **46** of the eMSP **28** framework, receives the signed contract data from the digital wallet **40** (e.g., via call/cps/forward/signedContractData). The contract data payload may include, for example: (1) the contract and the complete certificate chain, (2) the encrypted private key (and ephemeral key exchange data, e.g., Diffie-Hellman public parameters), (3) the PCID, and (4) the EMAID or other usage policies.

[0093] At step **726**, the CPS **46** confirms that the contract is authentic, adds a further eMSP signature or timestamp, and transmits it to the CCP **48**. In an embodiment, the CCP **48** may be configured to finalize, store, and distribute contract bundles—which may be XML-based (conforming to ISO 15118 or OCPP schemas) or in other machine-readable formats. In some embodiments, the CPS **46** could be merged with the CCP **48**. At step **728**, the CPS **46** returns a message to the digital wallet **40** indicating that the contract was successfully signed.

[0094] At step **730**, the CCP **48** generates a signed contract bundle. At step **732**, after the contract bundle is fully assembled, the CCP **48** notifies the digital wallet **40** that the contract is available for retrieval. At step **734**, the digital wallet **40**, using a secure transport protocol (such as TLS), calls an endpoint like GET/ccp/signedContractData?pcid=<PCID> and, at step **736**, downloads the finalized signed contract bundle. The bundle may include, for example, the signed leaf certificate, the Sub CA certificate, relevant signatures from the eMSP **28**, and any policy or rate tables mandated by ISO 15118 or OCPP.

[0095] At step **738**, the digital wallet **40** provisions the contract bundle to the EV **24**, which stores it in a secure storage module or hardware element, as described herein. Upon connecting to a CP **34** that trusts the SRC System Root CA (and thus also trusts CA Sub 1), the EV **24** automatically authenticates via the presented contract certificate. Because the EV **24** and the CP **34** share a recognized Root CA, the CP **34** can confirm the contract validity and use the user's linked DigitalCardID for billing and settlement through the SRC system, as described in detail below.

Binding Digital Card Id to Contract

[0096] FIG. **8** is a process flow diagram of a method **800** for binding a DigitalCardID (e.g. the DigitalCardID described in FIG. **6**) to the contract certificate (and certificate chain) generated by the eMSP **28**. At step **805**, the digital wallet **40** transmits a contract certificate request to the eMSP **28**. (See step **718** above for additional details.) At step **810**, the eMSP **28** returns a contract certificate to the digital wallet **40**. (See step **720** above for additional details.) The digital wallet **40** may be a digital wallet application executed by the user system **32**. In one embodiment, the user

system **32** may be integrated into the EV **24**. In alternative embodiments, the user system **32** may include a mobile computing device, such as a smartphone, tablet, laptop, and the like.

[0097] In some embodiments, if the digital wallet **40** cannot store different contracts for multiple DigitalCardIDs, the digital wallet **40** first needs to remove (or “unbind”) any previously active contracts before binding a new one. This ensures that the EV **24** is associated with only one valid contract at a time. Accordingly, at optional step **815**, the digital wallet **40** may initiate an unbind API call to the SRC system (e.g. the server system **30** shown in FIG. **1**). The digital wallet **40** may communicate over the API via a communications interface (e.g. communication interface **410** shown in FIG. **4**) of the user system **32**. In an example, the API call may include the HTTP Method/Endpoint “DELETE/profiles/appinstances.” The parameters may include, for example, and without limitation: (1) a serviceID, which may identify the eMSP service or subscription to unbind; (2) an srcClientID, which may include the digital wallet's SRC identifier; (3) the DigitalCardID, which may include the currently bound payment credential token; (4) a device identity, which may include a unique descriptor of the digital wallet device (e.g., serial number, hardware identifier, etc.); and (5) the contract (including the certificate chain up to CA Sub 1), which may include the existing contract certificate data to be unbound. In some embodiments, after the SRC system receives the unbind request, the SRC system may locate the binding record in a binding database (not shown) using the supplied parameters. The SRC system may remove and/or revoke the contract binding so that subsequent authentication or payment attempts under the old contract are no longer recognized. The SRC system may return an HTTP 200 or 204 success status to confirm that the contract has been unbound. If the contract is invalid or cannot be found, the SRC system may return an HTTP 404 or 400 error with diagnostic information.

[0098] At step **820**, after ensuring that any prior contract binding has been removed (if applicable), the digital wallet **40** establishes a secure session with the SRC system, and more particularly, the bind API. For example, in an embodiment, the digital wallet **40** may call the bind API using the HTTP Method/Endpoint “POST/profiles.” In certain embodiments, mutual TLS may be used, requiring both client and server certificate authentication. In addition, in other embodiments, the digital wallet **40** may typically supply credentials (e.g., OAuth tokens or an API key) to prove it is a registered digital card facilitator.

[0099] In the example embodiment, the API call parameters may include, for example, and without limitation: (1) a serviceID; (2) an srcClientID; (3) the DigitalCardID; (4) a device identity; and (5) the contract (including the certificate chain up to CA Sub 1). The contract may include, for example, the newly created or updated contract certificate for the EV **24**, signed by CA Sub 1 (which itself is under the SRC system's Root CA). The parameter may include the leaf certificate (representing the EV's public key and identity), any intermediate certificates up to the CA Sub 1, and any supporting policy or metadata, such as tariff details, expiration dates, or references to the user's EMAID.

[0100] At step **825**, the SRC system validates the contract. For example, in an embodiment, the SRC system may check the validity of the TLS session and verify that clientID matches a registered digital card facilitator. If mutual TLS is enabled, the SRC system may also examine the client-side certificate to confirm that it chains back to a trusted authority. The SRC system may also perform certificate chain verification steps. For example, the SRC system may parse the assurance data and reconstruct the certificate chain from the leaf certificate up to the CA Sub 1, which should ultimately reference the SRC System Root CA. In an example, the chain may be validated according to X.509 standards, checking expiration dates, signatures, and optionally performing revocation checks. The SRC system may perform a binding consistency check. For example, the SRC system may confirm that the DigitalCardID has not previously been bound to a different contract under the same serviceID (unless multiple bindings are allowed by policy). Similarly, the SRC may ensure that the device identity is associated with a recognized device.

[0101] At step **830**, the SRC system associates the contract with the DigitalCardID. For example,

after the certificate chain has been validated, the SRC system associates (binds) the DigitalCardID with the newly verified contract certificate in a binding database (not shown). The record or database entry ensures that subsequent charging or authentication attempts referencing the certificate chain will resolve to the specified DigitalCardID for billing and/or identity purposes. [0102] At step **835**, the SRC system responds to the digital wallet **40** with a success message (e.g., an HTTP 200 message) indicating that the contract binding is active. If an issue arises—such as an invalid certificate, an unrecognized device, or a conflict with an existing binding—the SRC system may reply with an error message (e.g., HTTP 400 or 409 message).

[0103] At this point, the digital wallet **40** and the EV **24** (once provisioned with this certificate bundle, as described above at step **738**) can engage in plug-and-charge transactions. As described below in detail, when the EV **24** presents the certificate chain at a cooperating CP **34** that trusts the SRC System Root CA, the CP **34** will validate the contract. The SRC system then facilitates transaction processing via the DigitalCardID, providing a frictionless charging experience without requiring manual payment input.

Contract-Based Payment Transaction Flow

[0104] FIG. **9** is a schematic diagram of the system **10** showing a data flow **900** among the parties during a payment transaction for charging an electric vehicle, such as the EV **24**. In the example embodiment, the payment transaction is a card-not-present (CNP) transaction where the consumer **22** initiates the payment transaction using, for example, and without limitation, account information of a payment account stored in the digital wallet **40** of the user system **32** to provide payment information associated with a cost of charging the EV **24**. In the example embodiment, the user system **32** may be integrated with the EV **24**. The user system **32** includes the digital wallet **40**.

[0105] In the example embodiment, the CP **34** includes a merchant computer device, such as the computing system **330** (shown in FIG. **3**). When the consumer **22** selects to initiate a transaction with the CP **34**, the consumer connects an electrical connector of the EV **24** (e.g. electrical connector **220** shown in FIG. **2**) with an electrical connector of the CP **34** (e.g. electrical connector **320** shown in FIG. **3**) to exchange handshaking information **902** as part of a handshaking procedure for establishing communication. The EV **24** and CP **34** establish a mutually authenticated and encrypted session through the handshaking procedure. This involves exchanging digital certificates for authentication and key exchange, resulting in an encrypted communication channel over a TCP/IP connection (e.g., over a TLS-encrypted session). It is noted that the messages between the EV **24** and the CP **34**, may, in at least some instances, conform to the International Organization for Standardization (ISO) Standard 15118, vehicle to grid communication interface, which is the ISO standard for systems that exchange electronic messages between an electric vehicle and an electric grid. In the example embodiment, the handshaking information **902** can be a plurality of ISO 15118 type messages.

[0106] In a preferred embodiment, the EV **24** and the CP **34** may exchange messages to confirm which ISO 15118 variant (or other protocol) they will use, for example, via

“SupportedAppProtocolReq/Res” message exchange. Once decided, the EV **24** may transmit a “SessionSetupReq” message to the CP **34**. The request may include an electric vehicle communication controller identifier (EVCCID) signed by the EV's private key, along with its EVCC certificate chain. The CP **34** may reply with a “SessionSetupRes” message, providing a unique SessionID, an electric vehicle supply equipment ID (EVSEID (i.e., the CP ID)), and the supply equipment communication controller (SECC) certificate chain. These steps ensure the EV **24** and CP **34** can authenticate one another via mutual TLS (mTLS) and that subsequent communication remains confidential and tamper-resistant. The handshaking information **902** received by the EV **24** from the CP **34** may also include merchant information. The merchant information may include a unique ID of the merchant **12** and/or other merchant information.

[0107] After secure communication is established between the CP **34** and the EV **24**, the EV **24** signals its intent to initiate payment for charging services. In response, the CP **34** provides a list of

supported payment services. The EV **24** then selects to pay via contract and generates authentication information **904**. The authentication information **904** includes, for example, a timestamp, a challenge response (e.g., returns a random number received from the CP **34**), and the contract (described above), and optionally, other verification information and/or merchant information. The EV **24** signs the authentication information **904** with a private key of the EV to encrypt the authentication information **904**.

[0108] If the ISO 15118-2 standard is selected for the session, in an example embodiment where the CP **34** supports contract-based payments (e.g., Plug-and-Charge (PnC)) and has the necessary root certificate authority for the SRC System (i.e., the server system **30**), the EV **24** may transmit a “ServiceDiscoveryReq” message to the CP **34** to inquire about available services at the CP **34**. The CP **34** may return a “ServiceDiscoveryRes” message containing a payment service list that identifies the supported payment methods. The list might include external identification means (EIM), contract-based payment options, or others depending on the CP's configuration. The EV **24** may select, for example, the contract-based option by sending a “PaymentServiceSelectionReq” message that specifies “SelectedPaymentOption=Contract” to the CP **34**. The CP **34** may respond with a “PaymentServiceSelectionRes” confirming that contract-based payment is permissible for the charging session. Before the EV **24** shares the contract details, the EV **24** must perform a consumer device cardholder verification method (CDCVM) if required for the transaction. The EV **24** may transmit a “PaymentDetailsReq” message containing the contract and certificate chain, and an electric mobility account identifier (EMAID). This data may prove that the EV **24** holds a valid contract tied to the EMAID. The CP **34** may return a “PaymentDetailsRes” message, providing a newly generated challenge (GenChallenge) for security. The EV **24**, having received the challenge, may sign the challenge locally using the private key of the contract. The signing action may be implicit in the transaction flow but critical for non-repudiation. The EV **24** may transmit an authorization request message back to the CP **34**, including the signed challenge, to prove that the EV **24** controls the contract and private key. The CP **34** may forward the authorization request to the merchant **12**. In an example, the authorization request may be carried via the open charge point protocol (OCPP) “AuthorizeRequest” message, which includes the contract certificate chain, the signed challenge data, and a timestamp.

[0109] In an alternative embodiment, when using the ISO 15118-20 standard, and if the CP **34** supports its advanced PnC features and recognizes the server system **30** (i.e., the SRC system) as an eMobility Service Provider (eMSP), the EV **24** may send an “AuthorizationSetupReq” message to the CP **34**. The message may include a session ID, a timestamp, and a digital signature. The CP **34** may respond with an “AuthorizationSetupRes” message that reaffirms the session details, lists the supported services (e.g., EIM, PnC), and includes a “PnC_ASResAuthorizationMode” parameter. The “PnC_ASResAuthorizationMode” parameter may include a random challenge number (UN); a roster of supported providers, among them the relevant SRC-based eMSP; and a request that the EV **24** provide its contract certificate chain. The message, including the challenge, is signed by the CP **34** and stored in “Authorization Data.” The EV **24** selects the contract to use based on one or more of the following: supported providers, default card contract, preferred contract for current merchant (i.e., CSO), and CDCVM status.

[0110] The EV **24** may transmit an authorization request message back to the CP **34**, referencing the session ID, a timestamp, and specifying the selected authorization service (PnC). The authorization response may also include the “PnC_AReqAuthorizationMode” parameter. The “PnC_AReqAuthorizationMode” parameter may include the random challenge number (UN) and the contract (signed by the SRC System and including the EV's EVCCID). The EV **24** may sign the payload with its own private key. The CP **34** may encapsulate the authorization request data into an OCPP “AuthorizeRequest” message. The “AuthorizeRequest” message includes the contract, the UN challenge, the timestamp, and an “IdToken” field to carry the EV's signature.

[0111] Referring back to FIG. **9**, in the depicted embodiment, regardless of whether ISO 15118-2

or ISO 15118-20 is used, the CP **34** provides the authentication information **904** to the merchant **12**. Optionally, the merchant **12** may forward the authentication information **904** to its acquirer **14**. [0112] In the exemplary embodiment, the merchant **12** or the acquirer **14** (1) extracts the DigitalCardID from the contract, (2) extracts the IdToken that includes the signed data from the EV, (3) creates a “DpaTransactionOptions” parameter (which might contain details about the transaction amount, pricing, or currency), and (4) creates a checkout request message with these components in the assurance data. The checkout request message is a structured message initiated by a digital payment application (DPA) to facilitate an online transaction.

[0113] The merchant **12** or acquirer **14** transmits the checkout request message **908** to the server system **30** (i.e., the SRC system). The checkout request message **908** includes the authentication information **904**, the extracted IdToken, and the extracted DigitalCardID. The server system **30** parses the checkout request message **908** to verify the authenticity and integrity of the payload. For example, the server system **30** verifies the EV's signature (e.g., using the EV's public key and via a trust chain) to confirm that the data (e.g., contract, challenge) has not been altered. The server system **30** also verifies the merchant's signature in a similar manner. The server system also verifies the DigitalCardID binding, i.e., that it corresponds to a valid payment credential (e.g., a tokenized card or PnC contract) recognized by the system. The server system **30** checks that the contract data matches what is registered for the EV **24** and that the contract has not been revoked or expired.

[0114] After validating the data elements received in the checkout request message and determining that the DigitalCardID is validly bound, the server system **30** may retrieve the associated payment credentials, such as the Primary Account Number (PAN) and other necessary card details, or generate a payment token and cryptogram for the transaction. The server system **30** then generates a checkout response message **910**. The checkout response message **910** may include the payment credentials (i.e., the PAN, token, or cryptogram) and a correlationID. The correlationID is a unique identifier created by the server system **30** to link all the events and messages related to the transaction. The server system **30** transmits the checkout response message **910** to the merchant **12** or acquirer **14**. If the binding validation fails—for example, if the DigitalCardID is expired, revoked, or does not match the EV identity in the contract—the server system **30** declines the request and returns an error or failure status indicating the reason for the mismatch.

[0115] In the example embodiment, after receiving the checkout response message **910**, the merchant **12** or acquirer **14** proceeds with processing a payment authorization. For example, the merchant **12** or acquirer **14** extracts payment credentials from the checkout response message **910** and sends an authorization request message **912** to the interchange network **16**. The authorization request message **912** may include the payment credentials and all necessary transaction details to process the payment transaction. The interchange network may enrich the authorization request message **912** as required and forward an enriched authorization request message **914** to the issuer **18**.

[0116] The issuer **18** processes the enriched authorization request message **914** to pre-authorize a payment amount included in transaction details. The issuer **18** parses the enriched authorization request message **914** to identify a PAN associated with the payment account of the consumer **22**. The issuer **18** subsequently determines whether the payment account includes sufficient available funds for processing the transaction. If the payment account includes sufficient funds to complete the transaction, the issuer **18** determines that authorization is successful and sends an authorization request response message **916** to the server system **30**. Otherwise, the issuer **18** determines that the authorization failed and sends an authorization failure message to the server system **30**.

[0117] Upon receiving the authorization request response message **916**, the server system **30** sends an authorization request response message **918** to the merchant **12** or acquirer **14**, indicating that the authorization was successful. If the acquirer **14** received the authorization request response message **918**, the acquirer **14** transmits an authorization request response message **920** to the merchant **12**. The authorization request response message **920** includes an indication of whether the

authorization was successful.

[0118] Upon receiving an authorization failure message, the server system **30** may send a failure message to user system **32**, indicating that the authorization was unsuccessful. Upon receiving the failure message, the consumer **22** may select a different payment method stored in the digital wallet **40** and submit new transaction information, including information identifying the different payment method.

[0119] If the authorization request response message **920** received by the merchant **12** indicates that the authorization was successful, the merchant **12** sends an instruction **922** to the CP **34** to begin charging the EV **24**. For example, in a preferred embodiment, the merchant **12** may transmit an OCPP “AuthorizeResponse” to the CP **34**. The response may include an “idTokenInfo” parameter (e.g., whether the authorization was accepted, blocked, expired, etc.) and may also include a “certificateStatus” parameter relevant to PnC-based contracts.

[0120] The CP **34** transmits an authorization response message **924** to the EV **24**, indicating the authorization was successful. For example, if using ISO 15118-2 (as discussed above), the CP **34** may send an “AuthorizationRes” message, typically containing a “ResponseCode” (e.g., accepted or rejected) and an “EVSEProcessing” parameter that indicates whether the EV **24** can immediately start charging or if the CP **34** requires more time to finalize setup. Alternatively, if using ISO 15118-20, the “AuthorizationRes” message may additionally include the SessionID, making the session reference explicit while still conveying the outcome and next steps for charging.

Push Remote Payment Transaction Flow

[0121] FIG. **10** is a schematic diagram of the system **10** showing an alternative data flow **1000** among the parties during a payment transaction for charging an EV. In this alternative embodiment, the payment transaction is a card-not-present (CNP) transaction where the consumer **22** initiates the payment transaction using, for example, and without limitation, account information of a payment account stored in the digital wallet **40** of the user system **32** to provide payment information associated with a cost of charging the EV **24**. In the example embodiment, the user system **32** may be integrated with the EV **24**. The user system **32** includes the digital wallet **40**.

[0122] In the example embodiment, the CP **34** includes a merchant computer device, such as the computing system **330** (shown in FIG. **3**). When the consumer **22** selects to initiate a transaction with the CP **34**, the consumer connects an electrical connector of the EV **24** (e.g. electrical connector **220** shown in FIG. **2**) with an electrical connector of the CP **34** (e.g. electrical connector **320** shown in FIG. **3**) to exchange handshaking information **1002** as part of a handshaking procedure for establishing communication. The EV **24** and CP **34** establish a mutually authenticated and encrypted session through the handshaking procedure. This involves exchanging digital certificates for authentication and key exchange, resulting in an encrypted communication channel over a TCP/IP connection (e.g., over a TLS-encrypted session). It is noted that the messages between the EV **24** and the CP **34**, may, in at least some instances, conform to the International Organization for Standardization (ISO) Standard 15118, vehicle to grid communication interface, which is the ISO standard for systems that exchange electronic messages between an electric vehicle and an electric grid. In the example embodiment, the handshaking information **1002** can be a plurality of ISO 15118 type messages.

[0123] In a preferred embodiment, the EV **24**, via the user system **32**, and the CP **34** may exchange messages to confirm which ISO 15118 variant (or other protocol) they will use, for example, via “SupportedAppProtocolReq/Res” message exchange. Once decided, the user system **32** may transmit a “SessionSetupReq” message to the CP **34**. The request may include an electric vehicle communication controller identifier (EVCCID) signed by the EV's private key, along with its EVCC certificate chain. The CP **34** transmits merchant information **1004** to the user system **32**. For example, the CP **34** may reply with a “SessionSetupRes” message, providing a unique SessionID, an electric vehicle supply equipment ID (EVSEID (i.e., the CP ID)), and the supply equipment communication controller (SECC) certificate chain including the CPO (i.e., merchant **12**)

certificate issuer ID. These steps ensure the EV **24** and CP **34** can authenticate one another via mutual TLS (mTLS) and that subsequent communication remains confidential and tamper-resistant. [0124] In some embodiments where the EV **24** or user system **32** is not connected to the Internet, the user system **32** may query the CP **34** to determine which external or value-added services are available (ServiceDiscoveryReq). The user system **32** may receive a ServiceDiscoveryRes from the CP **34** that enumerates possible services, including internet connectivity (Service ID=65). This step is performed for payment flows that rely on external networks, such as when the EV **24** does not have its own direct cellular or network link. This exchange of messages enables the user system **32** to learn specific parameters (e.g., ParameterSetID=4 for HTTPS over port **443**) for the service. The user system **32** then requests to activate the service (ServiceSelectionReq). After receiving the confirmation from the CP **34** (ServiceSelectionRes), the user system **32** can securely communicate with external systems over TLS via the CP's internet connection.

[0125] In the example embodiment, the user system **32** provides the merchant information **1004** to the digital wallet **40**. For example, user system **32** may select an appropriate key for signing the merchant information **1004**. The selection may depend on the consumer device cardholder verification method (CDCVM)—for example, if the consumer **22** has authenticated with the user system **32** via a biometric or PIN, the user system **32** may decide which private key to leverage. The user system **32** may compile a payload including its DigitalCardId (see FIG. **6**), a timestamp, the EVSEID, and the CPO Cert Issuer ID. The user system **32** may sign the payload with the selected private key and forward it to the digital wallet **40**.

[0126] It is noted that the user system **32** has a DigitalCardId. During a period prior to initiating the payment transaction for charging an EV **24**, the user system **32** may establish a secure session with the server system **30** (i.e., the SRC system), and more particularly, a bind API, to bind the DigitalCardID with the public key of the user system **32**. For example, in an embodiment, the user system **32** may call the bind API using the HTTP Method/Endpoint “POST/profiles.” The API call parameters may include, for example, and without limitation: (1) a serviceID; (2) an srcClientID; (3) the DigitalCardID; (4) a device identity; and (5) the public key of the user system **32**. The server system **30** may associate the public key with the DigitalCardID. For example, the server system **30** associates (binds) the DigitalCardID with the public key in a binding database (not shown). The record or database entry ensures that subsequent charging or authentication attempts signed by the public key will resolve to the specified DigitalCardID for billing and/or identity purposes.

[0127] Referring back to FIG. **10**, the digital wallet **40** initiates communication with the server system **30** via API **1006**. The digital wallet **40** provides the merchant information **1004** and signature to the server system **30** of the interchange network **16** via the API **1006**. The server system **30** verifies the merchant information **1004**. Such verification includes verifying that the EVSEID and CPO Cert Issuer ID correspond to recognized IDs in the system (DPA_ID and SRCIP_ID), fetching the EV's public key based on the DigitalCardId, and confirming the authenticity of the signature. Verifying the information may include querying the database **26**.

[0128] After verifying the information **1004**, the server system **30** may retrieve the associated payment credentials, such as the Primary Account Number (PAN) and other necessary card details, or generate a payment token and cryptogram for the transaction. The server system **30** then generates a payload, including the payment credentials (i.e., the PAN, token, or cryptogram) and other merchant data. The server system **30** may send the payload to the acquirer **14**. The acquirer **14** may validate the payload and send an authorization request **1007** to the interchange network **16**. The authorization request **1007** may include the payload. The interchange network **16** may generate transaction information **1008** based on the authorization request **1007**. The transaction information **1008** may include the payload. The interchange network **16** sends the transaction information **1008** to the issuer **18**.

[0129] The issuer **18** processes the transaction information **1008** to pre-authorize a payment

amount included in the transaction information **1008**. The issuer **18** parses the transaction information **1008** to identify a PAN associated with the payment account of the consumer **22**. The issuer **18** subsequently determines whether the payment account includes sufficient available funds for processing the transaction. If the payment account includes sufficient funds to complete the transaction, the issuer **18** determines that pre-authorization is successful and sends a pre-authorization success message to the interchange network **16**. Otherwise, the issuer **18** determines that the pre-authorization has failed and sends a pre-authorization failure message to the interchange network **16**. Upon receiving the pre-authorization success message, the interchange network **16** sends a message **1009** to the acquirer **14** indicating that the pre-authorization was successful. The acquirer **14** sends a message **1010** to the merchant **12**. The message **1010** includes the unique identifier of the CP **34** and an indication of whether the pre-authorization was successful.

[0130] Upon receiving the pre-authorization failure message, the interchange network **16** sends a failure message to user system **32** indicating that the pre-authorization was not successful. Upon receiving the failure message, the consumer **22** may select a different payment method stored in the digital wallet **40** and re-submit the transaction information **1008**, including information identifying the different payment method.

[0131] The user system **32** sends an authorization request **1011** to the CP **34**, which prompts the CP **34** to ask the merchant **12** for an updated authorization status message **1014**. Upon receiving an "OK" message **1014** from the merchant **12**, the CP **34** returns an authorization response "OK" message **1016** to the EV **24**, confirming that the charging session may proceed. This asynchronous loop may be required due to the potential latency in online payment authorization, ensuring that the EV **24** only commences charging after the CP **34** has retrieved the payment authorization result from the interchange network **16**. After charging is completed, the CP **34** sends a report to the merchant **12**, indicating a final amount to charge the payment account of the consumer **22**.

Computer-Implemented Methods for Ev Charging Payment Transaction

[0132] FIG. **11** is a schematic diagram of the system **10** showing an alternative data flow **1100** among the parties during a payment transaction for charging an EV. In this alternative embodiment, the payment transaction is a card-not-present transaction where the consumer **22** initiates the payment transaction using, for example, and without limitation, account information stored in the digital wallet **40** of the user system **32** to provide payment information associated with a cost of charging the EV **24**.

[0133] In the example embodiment, the user system **32** retrieves payment account information **1102**, such as a PAN or DigitalCardID, from the digital wallet **40**. The payment account information **1102** is sent to the server system **30** of the interchange network **16** to initiate a registration process for registering the payment information **1102** with the interchange network **16**. The server system **30** processes the payment information **1102** and stores a profile of the consumer **22**, including the payment information **1102**, in the database **26**. The user system **32** generates a public/private key pair (i.e., and asymmetric key pair) associated with an EV unique identifier (ID) of the EV **24**. In the example embodiment, the unique identifier of the EV **24** may be a vehicle identification number (VIN). The private key is stored in the digital wallet **40**. The digital wallet **40** binds the public key to the EV unique ID and DigitalCardID to generate a bound EV ID **1104**. The bound EV ID **1104** is sent to the server system **30**. The server system **30** stores the bound EV ID **1104** in the profile of the consumer **22** in the database **26** to complete the registration process.

[0134] In the example embodiment, the CP **34** includes a merchant computer device (not shown), such as the computing system **330** (shown in FIG. **3**). To initiate the EV charging process, the consumer **22** connects an electrical connector of the EV **24** (e.g. electrical connector **220** shown in FIG. **2**) with an electrical connector of the CP **34** (e.g. electrical connector **320** shown in FIG. **3**) to exchange handshaking information **1106** as part of a handshaking procedure for establishing communication between the EV **24** and the CP **34**. It is noted that the messages between the EV **24**

and the CP **34**, may, in at least some instances, conform to the International Organization for Standardization (ISO) Standard 15118, vehicle to grid communication interface, which is the ISO standard for systems that exchange electronic messages between an electric vehicle and an electric grid. In the example embodiment, the handshaking information **1106** can be a plurality of ISO 15118 type messages.

[0135] After communication is established between the CP **34** and the EV **24**, the CP **34** transmits merchant information **1108** to the EV **24**. The merchant information **1108** includes a unique identifier of the CP **34**, a unique identifier of the merchant **12**, and other merchant information such as an estimated cost of charging the EV **24** and a timestamp. The user system **32** provides the merchant information **1108** to the digital wallet **40**. The digital wallet **40** signs the merchant information **1110** with the private key of the public/private key pair to generate signed merchant information **1110** that includes the EV ID **1104**. The digital wallet **40** provides the signed merchant information **1110** to the CP **34**.

[0136] In response to receiving the signed merchant information **1110**, the CP **34** sends a charge request **1112**, including the signed merchant information **1110**, to the merchant **12**. At **1114**, the merchant **12** forwards the charge request **1112** to the acquirer **14**. The acquirer **14** initiates a communication session with interchange API **1116** to authorize the charge request **1112** with the server system **30**. The acquirer **14** forwards the charge request **1112** to the server system **30** via the interchange API **1116a**.

[0137] The server system **30** decrypts and verifies the unique IDs for the CP **34**, the EV **24**, and the merchant **12**. Verifying the unique IDs may include querying the database **26** and determining whether any stored IDs match the unique identifiers of the CP **34** and the merchant **12**. In response to verifying the unique IDs, the server system **30** retrieves a DigitalCardID (e.g. the DigitalCardID described in reference to FIG. **6**) from the database **26**. The DigitalCardID is associated with a payment card registered with the server system **30**.

[0138] After retrieving the DigitalCardID, the server system **30** generates a payload. The payload may include the DigitalCardID; the unique IDs of the CP **34**, the merchant **12**, and the EV; and other merchant data. The server system **30** may send the payload to the acquirer **14** via the API **1116a**. The acquirer **14** may validate the payload and send an authentication request **1116b** to the server **30**. The authentication request **1116b** may include the payload. The server system **30** verifies the unique IDs of the EV, the CP, and the merchant, which are included in the payload of the authentication request **1116b**. The server **30** may generate transaction information **1118** based on the authorization request **1116b**. The transaction information **1118** may include the payload. The server system **30** sends the transaction information **1118** to the issuer **18**.

[0139] The issuer **18** processes the transaction information **1118** to pre-authorize a payment amount included in the transaction information. The issuer **18** parses the transaction information **1118** to identify a PAN associated with the payment account of the consumer **22**. The issuer **18** subsequently determines whether the payment account includes sufficient available funds for processing the transaction. If the payment account includes sufficient funds to complete the transaction, the issuer **18** determines that pre-authorization is successful and sends a pre-authorization success message to the server system. Otherwise, the issuer **18** determines that the pre-authorization has failed and sends a pre-authorization failure message to the server system **30**. Upon receiving the pre-authorization success message, the server system **30** sends a message **1119** to the acquirer **14** indicating that the pre-authorization was successful. The acquirer **14** sends a message **1120** to the merchant **12**. The message **1120** includes the unique ID of the CP **34** and an indication of whether the pre-authorization was successful.

[0140] Upon receiving the pre-authorization failure message, the server system **30** sends a failure message to user system **32** indicating that the pre-authorization was not successful. Upon receiving the failure message, the consumer **22** may select a different payment method stored in the digital wallet **40** and re-submit the transaction information **1118**, including information identifying the

different payment method.

[0141] If the message **1120** received by the merchant **12** indicates that pre-authorization was successful, the merchant **12** sends an instruction **1122** to the CP **34** to begin charging the EV **24**. After charging is completed, the CP **34** sends a report to the merchant **12** indicating a final amount to charge the payment account of the consumer **22**.

[0142] FIG. **12** is a process flow diagram of a method **1200** for charging an electric vehicle (e.g. the EV **24** shown in FIG. **1**). At step **1205** the EV is connected to a CP (e.g. the CP **34** shown in FIG. **1**) via a charging cable (e.g. charging cable **310** shown in FIG. **3**). At step **1210**, the EV queries the CP and receives an indication of supported communication protocols from the CP. In one embodiment, the supported communication protocols include various protocols through which payment may be provided. At step **1215**, the EV and CP perform a handshaking process (as described in reference to FIGS. **8-10**) to establish electronic communication with one another. As part of the handshaking process, the EV provides an EV ID (e.g. the VIN of the EV) to the CP, and the CP provides a CP ID and a merchant ID to the EV. After communication is established, at step **1220**, the EV provides authorization information to the CP. The authorization information includes a timestamp, the EV ID, the CP ID, the merchant ID, and other information used for authorization. The authorization information is signed by the EV using a public key (e.g. the public key described in reference to FIG. **7**). The CP provides the authorization information to the merchant. The merchant may provide the authorization information to the server system.

[0143] At step **1225**, the authorization information is validated by the server system. Validating the transaction information by the server system includes parsing the authorization information to identify the EV ID. The server system uses the EV ID to query a database (e.g. the database **26** shown in FIG. **1**) for retrieving a public key of the vehicle (e.g. the public key described in reference to FIG. **7**). The server system validates the authorization information by decrypting the signed authorization information using the EV public key. At step **1230**, the server system may retrieve transaction information in response to validating the authorization information. The server system retrieves a DigitalCardID (e.g. the DigitalCardID described in reference to FIG. **6**) from a database (e.g. the database **26** shown in FIG. **1**). The DigitalCardID is associated with a payment card registered with the server system. The server system may send the transaction information including the DigitalCardID to the acquirer. The acquirer validates the transaction information and sends the transaction information and an authorization request back to the server system.

[0144] At step **1235**, the server system authorizes the transaction information and the authorization request received from the acquirer. The server system sends an indication to the merchant that the authorization was successful. In response to receiving the indication, the merchant instructs the CP to begin charging the EV. At step **1240**, the CP begins providing a charging current to the EV.

[0145] FIGS. **13A** and **13B** are process flow diagrams of a method **1300** for charging an electric vehicle (e.g. the EV **24** shown in FIG. **1**). At step **1310**, registration information is received by a server system (e.g. the server system **30** shown in FIG. **1**) from an acquirer (e.g. the acquirer **14** shown in FIG. **1**) and a CPO (e.g. the merchant **12** shown in FIG. **1**). The registration information includes information associated with various entities to be registered with the server system, including one or more CPs (e.g. the CPs **34** shown in FIG. **1**) and one or more CPOs. At step **1315**, the server system **30** processes the registration information and assigns respective unique acquirer identifiers (ID) to the acquirer and the CPO. At step **1320**, the server system **30** assigns respective unique CP IDs to each of the one or more CPs.

[0146] At step **1325**, a call is established with an API (e.g. the API **906** shown in FIG. **9**), to register one or more EVs (e.g. the EV **24** shown in FIG. **1**) with the server system **30**. The EV may send a unique EV ID to the server system as part of the registration process. At step **1330**, the server system assigns a unique DigitalCardID to the EV. At step **1335**, the EV stores the DigitalCardID in a digital wallet (e.g. the digital wallet **40** shown in FIG. **1**). The DigitalCardID may identify a payment account associated with the user.

[0147] In the exemplary embodiment, the unique IDs for the acquirer, the CPO, the one or more CPs, and the one or more EVs may include previously existing IDs provided by the respective entities. For example, the unique EV ID may comprise a VIN associated with the EV In an alternative embodiment, the unique IDs may be generated by the server system using any manner known to one of ordinary skill in the art.

[0148] At step **1345**, the EV is connected to a CP via a charging cable (e.g. charging cable **310** shown in FIG. **3**). At step **1350**, the EV and CP perform a handshaking process (as described in reference to FIGS. **9-11**) to establish electronic communication with one another. After communication is established, at step **1355**, the CP sends merchant information to the EV The merchant information includes the CP ID, the acquirer ID, the CPO ID, and other merchant information.

[0149] At step **1360**, the CP determines whether the EV is connected to a wireless communication network (e.g. network **20** shown in FIG. **1**). If the EV is connected to the wireless network, the method proceeds to step **1365**. At step **1365**, the EV establishes communication with the API to provide transaction information to the server system. The transaction information includes the EV ID, the acquirer ID, the CP ID, the CPO ID, and the other merchant information. The other merchant information may include a request amount to charge a payment account of the consumer **22** and a timestamp received from the CP **34**.

[0150] At step **1370**, if the EV **24** is not connected to the wireless network, the CP **34** provides an indication to the EV **24** that the CP **34** supports a value-added service (VAS). The VAS provides remote communication capabilities to the EV **24** through the CP **34** by utilizing network connections of the CP **34** to provide the transaction information to the server system.

[0151] At step **1375**, the server system authorizes the transaction information by validating the EV ID, the acquirer ID, the CP ID, the CPO ID, and the other merchant information. Validating the transaction information by the server system includes parsing the transaction information to identify the EV ID. The server system uses the EV ID to query a database (e.g. the database **26** shown in FIG. **1**) for retrieving a public key of the vehicle (e.g. the public key described in reference to FIG. **7**). The DigitalCardID is associated with a payment card registered with the server system. The server system may send the transaction information including the DigitalCardID to the acquirer. The acquirer may validate the transaction information and send the transaction information and an authorization request back to the server system.

[0152] The server system authorizes the transaction information and the authorization request received from the acquirer. The server system sends an indication to the merchant that the authorization was successful. In response to receiving the indication, the merchant instructs the CP to begin charging the EV As described in FIG. **10**, the server system sends an indication that the authorization is successful. At step **1380**, the CP begins providing a charging current to the EV.

[0153] FIGS. **14A** and **14B** are process flow diagrams of an alternative process **1400** for charging an EV, such as the EV **24** (shown in FIG. **1**). At step **1405**, registration information is received by a server system (e.g. the server system **30** shown in FIG. **1**) from an acquirer (e.g. the acquirer **14** shown in FIG. **1**) and a CPO (e.g. the merchant **12** shown in FIG. **1**). The registration information includes information associated with various entities to be registered with the server system, including one or more CPs (e.g. the CPs **34** shown in FIG. **1**) and one or more CPOs. At step **1410**, the server system **30** processes the registration information and assigns respective unique acquirer identifiers (ID) to the acquirer and the CPO. At step **1415**, the server system **30** assigns unique CP IDs to each of the one or more charge points.

[0154] At step **1420**, a digital wallet (e.g. digital wallet **40** shown in FIG. **1**) of an EV (e.g. EV **24** shown in FIG. **1**) generates a public/private key pair (i.e., an asymmetric key pair) to encrypt information associated with charging the EV. At step **1425**, an API call is established to register the EV with the server system. At step **1430**, the server system receives registration data from the EV. The registration data includes the public key of the public/private key pair. At step **1435**, the server

system assigns a unique DigitalCardID to the EV. At step **1440**, the DigitalCardID is stored in the digital wallet.

[0155] At step **1450**, the EV is connected to a CP via a charging cable (e.g. charging cable **310** shown in FIG. **3**). At step **1455**, the EV and CP perform a handshaking process (as described in reference to FIGS. **9-11**) to establish electronic communication with one another. In one embodiment, the CP sends a message to the EV indicating that the CP supports a value-added service (VAS) for retrieving payment information from the server system. The VAS includes instructions for structuring a data packet used to communicate payment information. After communication is established, at step **1460**, the CP sends merchant information to the EV. The merchant information includes the CP ID, the acquirer ID, the CPO ID, and other merchant information.

[0156] At step **1465**, the user system of the EV launches a digital wallet application to access the digital wallet and signs the merchant information using the private key of the public/private key pair. The EV subsequently sends the signed merchant information and the EV ID to the CP. In one embodiment, the EV sends the signed merchant information to the CP by overloading a data packet with payment information used by the server system to retrieve payment account information associated with the user. In an alternative embodiment, the EV sends the signed merchant information to the CP by generating a data packet in accordance with the VAS provided by the CP. At step **1470**, the signed merchant information is sent to the acquirer by the CPO associated with the CP. At step **1475**, the acquirer sends the signed merchant information to the server system to be validated. The server system validates the signed merchant information by decrypting the signed merchant information using the public key of the EV.

[0157] At step **1480**, the server system authorizes the transaction information by validating the EV ID, the acquirer ID, the CP ID, the CPO ID, and the other merchant information. Validating the transaction information by the server system includes parsing the transaction information to identify the EV ID. The server system uses the EV ID to query a database (e.g. the database **26** shown in FIG. **1**) for retrieving a public key of the vehicle (e.g. the public key described in reference to FIG. **7**). The DigitalCardID is associated with a payment card registered with the server system. The server system may send the transaction information including the DigitalCardID to the acquirer. The acquirer may validate the transaction information and send the transaction information and an authorization request back to the server system.

[0158] The server system sends the transaction information and the authorization request to an issuer (e.g. the issuer **18**, as described in FIG. **7**). The issuer may authorize the transaction information and the authorization request. The issuer may send an indication of whether the transaction information and authorization request are successfully authorized. The server system sends an indication to the merchant that the authorization was successful. In response to receiving the indication, the merchant instructs the CP to begin charging the EV. At step **1485**, the CP begins providing a charging current to the EV

Parallel Session Payment Transaction Flow

[0159] FIGS. **15A** and **15B** are process flow diagrams of a parallel session process **1500** for charging an EV, such as the EV **24** (shown in FIG. **1**). At step **1502**, registration information is received by the server system **30** (shown in FIG. **1**) from the acquirer **14** (shown in FIG. **1**) and the CPO or merchant **12** (shown in FIG. **1**). The registration information includes information associated with various entities to be registered with the server system **30**, including one or more CPs **34** (shown in FIG. **1**). At step **1504**, the server system **30** processes the registration information and assigns respective unique acquirer identifiers (IDs) to the acquirer **14** and the merchant **12**. At step **1506**, the server system **30** assigns unique CP IDs to each of the one or more CPs **34**.

[0160] At step **1508**, one of the user system **32** or digital wallet **40** (shown in FIG. **1**) of the EV **24** generates a public/private key pair (i.e., an asymmetric key pair) to encrypt information associated with charging the EV **24**. If the digital wallet **40** generates the key pair, the digital wallet **40**

securely provisions the private key within the EV (e.g., within the user system **32**). If user system **32** generates the key pair, the user system **32** shares the public key with digital wallet **40**.

[0161] It is noted that the user system **32** or digital wallet **40** has a DigitalCardID stored thereon (described above). At step **1510**, the user system **32** may establish a secure session with the server system **30** (i.e., the SRC system), and more particularly, a bind API, to bind the DigitalCardID with the public key of the user system **32**. For example, in an embodiment, the user system **32** may call the bind API using the HTTP Method/Endpoint “POST/profiles.” The API call parameters may include, for example, and without limitation: (1) a serviceID; (2) an srcClientID; (3) the DigitalCardID; (4) a device identity; and (5) the public key of the user system **32**.

[0162] At step **1512**, the server system **30** receives the API call and associated data elements from the user system **32**. The registration data includes the public key of the public/private key pair and the DigitalCardID. At step **1514**, the server system **30** associates (binds) the public key with the DigitalCardID. At step **1516**, the server system **30** stores the DigitalCardID-to-public key binding in a database (not shown). The record or database entry ensures that subsequent charging or authentication attempts signed by the public key will resolve to the specified DigitalCardID for billing and/or identity purposes.

[0163] At step **1518**, the EV **24** is connected to a CP via a charging cable (e.g. charging cable **310** shown in FIG. **3**). At step **1520**, the EV and CP perform a handshaking process (as described in reference to FIG. **9**) to establish electronic communication with each other. In the exemplary embodiment, a transport control protocol/internet protocol (TCP/IP) session is established (e.g., over a TLS-encrypted session). The TCP/IP provides a secure parallel communication session such that transaction information is communicated separately from communication channels used to exchange EV charging information. In one embodiment, the TCP/IP session facilitates communication of ISO 15118 type messages. More specifically, the TCP/IP session may utilize an ISO 15118-20 communication channel for exchanging ISO 15118 type messages.

[0164] After secure communication is established between the CP **34** and the EV **24**, at step **1522**, the EV **24** signals its intent to initiate charging and payment for the charging services. In response, at step **1524**, the CP **34** sends transaction information to the EV **24** through the TCP/IP session. The transaction information includes the CP ID, the acquirer ID, the CPO-ID, and other transaction information. More particularly, the CP **34** may provide a list of supported payment services. The EV **24** may select to pay via the DigitalCardID. The CP **34** may then send challenge data to the EV **24**.

[0165] At step **1526**, the user system **32** of the EV **24** launches the digital wallet **40** and signs the transaction information using the private key of the public/private key pair. The EV **24** subsequently sends the signed transaction information and the DigitalCardID to the CP **34**. In one embodiment, the EV **24** sends the signed transaction information to the CP **34** by overloading a data packet (IdToken) with payment information used by the server system **30** to retrieve payment account information associated with the user. The signed transaction information may include, for example, the DigitalCardID, a timestamp, the EVSEID, and the CPO Cert Issuer ID (Described herein in reference to FIG. **10**).

[0166] At step **1528**, the CP **34** provides the signed transaction information to the merchant **12**. Optionally, the merchant **12** may forward the authentication information **904** to its acquirer **14**. In the exemplary embodiment, the merchant **12** or the acquirer **14** (1) extracts the DigitalCardID, (2) extracts the IdToken that includes the signed data from the EV **24**, (3) creates a “DpaTransactionOptions” parameter (which might contain details about the transaction amount, pricing, or currency), and (4) creates a checkout request message with these components in the assurance data.

[0167] At step **1530**, the merchant **12** or acquirer **14** transmits the checkout request message to the server system **30** (i.e., the SRC system). The checkout request message includes the authentication information, the extracted IdToken, and the extracted DigitalCardID. The server system **30** parses

the checkout request message to verify the authenticity and integrity of the payload. For example, at step **1532**, the server system **30** verifies the merchant information. Verification of the merchant information includes verifying that the EVSEID and CPO Cert Issuer ID correspond to recognized IDs in the system (DPA_ID and SRCIP_ID), fetching the EV's public key based on the DigitalCardId, and confirming the authenticity of the signature.

[0168] After validating the data elements received in the checkout request message and determining that the DigitalCardID is validly bound, at step **1534**, the server system **30** authorizes the transaction data. In particular, the server system **30** may retrieve the associated payment credentials, such as the Primary Account Number (PAN) and other necessary card details, or generate a payment token and cryptogram for the transaction. The server system **30** may generate a checkout response message. The checkout response message may include the payment credentials (i.e., the PAN, token, or cryptogram) and a correlationID. The server system **30** may transmit the checkout response message to the merchant **12** or acquirer **14**. If the binding validation fails—for example, if the DigitalCardID is expired, revoked, or does not match the EV identity in the contract—the server system **30** declines the request and returns an error or failure status indicating the reason for the mismatch.

[0169] In the example embodiment, after receiving the checkout response message, the merchant **12** or acquirer **14** proceeds with processing a payment authorization at step **1536** (as described further herein with respect to FIG. **10**). At step **1538**, the merchant **12** or acquirer **14** transmits the authorization result to the CP **34** for storage thereon.

[0170] Substantially simultaneously or in parallel with step **1524**, at step **1540**, the CP **34** responds to the intent to initiate charging services message from the EV **24** with an authorization setup message. For example, the CP **34** may reply to the EV **24** with a “AuthorizationSetupRes” message, providing a unique Session ID, a list of the supported services (e.g., EIM, PnC) signed with the supply equipment communication controller (SECC) certificate, and the SECC certificate chain including the CPO certificate issuer ID, similarly signed.

[0171] At step **1542**, the EV **24** may transmit an authorization request message back to the CP **34**, referencing the session ID, a timestamp, and specifying the selected authorization service (EIM). At step **1544**, the CP **34** checks for the authorization result (described above in step **1538**). Notably, the CP **34** waits until a positive authorization result is provided by the merchant **12** before proceeding with charging of the EV **24**.

[0172] After receiving a positive authorization result, at step **1546**, the CP **34** transmits an authorization response message to the EV **24**, indicating the authorization was successful. As noted herein, in an embodiment using ISO 15118-20, the CP **34** may send an “AuthorizationRes” message, typically containing (1) a “ResponseCode” (e.g., accepted or rejected), (2) an “EVSEProcessing” parameter that indicates whether the EV **24** can immediately start charging or if the CP **34** requires more time to finalize setup, and (3) the Session ID, making the session reference explicit while still conveying the outcome and next steps for charging. At step **1548**, the CP **34** provides charging current to the EV **24** via the charging cable **310**. It is noted that steps **1524-1538** are performed substantially in parallel with steps **1540-1544** of this parallel session process **1500**.
Computer-Implemented Method for Charging an EV

[0173] FIG. **16** is a schematic diagram of the system **10** showing a data flow **1600** among the parties during a payment transaction for charging an EV. In this alternative embodiment, the payment transaction is a card-not-present transaction where the consumer **22** initiates the payment transaction using, for example, and without limitation, account information stored in the digital wallet **40** of the user system **32** to provide payment information associated with a cost of charging the EV **24**.

[0174] In the example embodiment, the user system **32** retrieves payment account information **1602**, such as a PAN or DigitalCardID, from the digital wallet **40**. The payment account information **1602** is sent to the server system **30** of the interchange network **16** to initiate a

registration process for registering the payment information **1602** with the interchange network **16**. The server system **30** processes the payment information **1602** and stores a profile of the consumer **22**, including the payment information **1602**, in the database **26**. The user system **32** generates a public/private key pair associated with an EV unique identifier (ID) of the EV **24**. In the example embodiment, the unique identifier of the EV **24** may be a vehicle identification number (VIN). The private key is stored in the digital wallet **40**. The digital wallet **40** binds the public key to the EV unique ID and DigitalCardID to generate a bound EV ID **1604**. The bound EV ID **1604** is sent to the server system **30**. The server system **30** stores the bound EV ID **1604** in the profile of the consumer **22** in the database **26** to complete the registration process.

[0175] In the example embodiment, the CP **34** includes a merchant computer device (not shown), such as the computing system **330** (shown in FIG. 3). To initiate the EV charging process, the consumer **22** connects an electrical connector of the EV **24** (e.g. electrical connector **220** shown in FIG. 2) with an electrical connector of the CP **34** (e.g. electrical connector **320** shown in FIG. 3) to exchange handshaking information **1606** as part of a handshaking procedure for establishing communication between the EV **24** and the CP **34**. It is noted that the messages between the EV **24** and the CP **34**, may, in at least some instances, conform to the International Organization for Standardization (ISO) Standard 15118, vehicle to grid communication interface, which is the ISO standard for systems that exchange electronic messages between an electric vehicle and an electric grid. In the example embodiment, the handshaking information **1606** can be a plurality of ISO 15118 type messages.

[0176] After communication is established between the CP **34** and the EV **24**, the CP **34** transmits merchant information **1608** to the EV **24**. The merchant information **1608** includes a unique identifier of the CP **34**, a unique identifier of the merchant **12**, and other merchant information such as an estimated cost of charging the EV **24** and a timestamp. The user system **32** provides the merchant information **1608** to the digital wallet **40**. The digital wallet **40** signs the merchant information **1610** with the private key of the public/private key pair to generate signed merchant information **1610** that includes the EV ID **1604**. The digital wallet **40** provides the signed merchant information **1610** to the CP **34**.

[0177] In response to receiving the signed merchant information **1610**, the CP **34** sends a charge request **1612**, including the signed merchant information **1610**, to the merchant **12**. At **1614**, the merchant **12** forwards the charge request **1612** to the server system **30**. The server system **30** initiates a communication session with interchange API **1616** to authorize the charge request **1612** with the acquirer **14**. The acquirer **14** forwards the charge request **1612** to the server system **30** via the interchange API **1616a**.

[0178] The server system **30** decrypts and verifies the unique IDs for the CP **34**, the EV **24**, and the merchant **12**. Verifying the unique IDs may include querying the database **26** and determining whether any stored IDs match the unique identifiers of the CP **34** and the merchant **12**. In response to verifying the unique IDs, the server system **30** retrieves a DigitalCardID (e.g. the DigitalCardID described in reference to FIG. 6) from the database **26**. The DigitalCardID is associated with a payment card registered with the server system **30**.

[0179] After retrieving the DigitalCardID, the server system **30** generates a payload. The payload may include the DigitalCardID; the unique IDs of the CP **34**, the merchant **12**, and the EV; and other merchant data. The server system **30** may send the payload to the acquirer **14** via the API **1616a**. The acquirer **14** may validate the payload and send an authentication request **1616b** to the server **30**. The authentication request **1616b** may include the payload. The server system **30** verifies the unique IDs of the EV, the CP, and the merchant, which are included in the payload of the authentication request **1616b**. The server **30** may generate transaction information **1618** based on the authorization request **1616b**. The transaction information **1618** may include the payload. The server system **30** sends the transaction information **1618** to the issuer **18**.

[0180] The issuer **18** processes the transaction information **1618** to pre-authorize a payment

amount included in the transaction information. The issuer **18** parses the transaction information **1618** to identify a PAN associated with the payment account of the consumer **22**. The issuer **18** subsequently determines whether the payment account includes sufficient available funds for processing the transaction. If the payment account includes sufficient funds to complete the transaction, the issuer **18** determines that pre-authorization is successful and sends a pre-authorization success message to the server system. Otherwise, the issuer **18** determines that the pre-authorization has failed and sends a pre-authorization failure message to the server system **30**. Upon receiving the pre-authorization success message, the server system **30** sends a message **1619** to the merchant **12** indicating that the pre-authorization was successful. The message may include an encrypted payload including payment information for charging the EV.

[0181] Upon receiving the pre-authorization failure message, the server system **30** sends a failure message to user system **32** indicating that the pre-authorization was not successful. Upon receiving the failure message, the consumer **22** may select a different payment method stored in the digital wallet **40** and re-submit the transaction information **1618**, including information identifying the different payment method.

[0182] The merchant **12** may decrypt the payload. In an example, the payload may be encrypted with a public key of the merchant **12**. The merchant **12** may decrypt the payload using a private key of the merchant **12**. In another example, the payload may be encrypted with a private key of the acquirer **14**. The merchant **12** may send the payload to the acquirer **14** such that the acquirer may decrypt the payload using a public key of the acquirer **14**. Alternatively, the merchant **12** may have access to the public key of the acquirer **14** and may decrypt the payload using the public key of the acquirer **14**. In another example, the acquirer **14** may receive a correlation ID instead of the payload. The merchant **12** may query an API of the server system **30** to retrieve the payload. In this example, the payload may be encrypted with the private key of the merchant **12** or the private key of the acquirer **14**.

[0183] If the message **1619** received by the merchant **12** indicates that pre-authorization was successful, the merchant **12** sends an instruction **1620** to the CP **34** to begin charging the EV **24**. After charging is completed, the CP **34** sends a report to the merchant **12** indicating a final amount to charge the payment account of the consumer **22**.

ADDITIONAL CONSIDERATIONS

[0184] In this description, references to “one embodiment,” “an embodiment,” or “embodiments” mean that the feature or features being referred to are included in at least one embodiment of the technology. Separate references to “one embodiment,” “an embodiment,” or “embodiments” in this description do not necessarily refer to the same embodiment and are also not mutually exclusive unless so stated and/or except as will be readily apparent to those skilled in the art from the description. For example, a feature, structure, act, etc. described in one embodiment may also be included in other embodiments but is not necessarily included. Thus, the current technology can include a variety of combinations and/or integrations of the embodiments described herein.

[0185] The detailed description is to be construed as exemplary only and does not describe every possible embodiment because describing every possible embodiment would be impractical. Numerous alternative embodiments may be implemented, using either current technology or technology developed after the filing date of this application, which would still fall within the scope of the invention.

[0186] Throughout this specification, plural instances may implement components, operations, or structures described as a single instance. Although individual operations of one or more methods are illustrated and described as separate operations, one or more of the individual operations may be performed concurrently, and nothing requires that the operations be performed in the order recited or illustrated. Structures and functionality presented as separate components in example configurations may be implemented as a combined structure or component. Similarly, structures and functionality presented as a single component may be implemented as separate components.

These and other variations, modifications, additions, and improvements fall within the scope of the subject matter herein. The foregoing statements in this paragraph shall apply unless so stated in the description and/or except as will be readily apparent to those skilled in the art from the description. [0187] As used herein, the term “database” includes either a body of data, a relational database management system (RDBMS), or both. As used herein, a database includes, for example, and without limitation, a collection of data including hierarchical databases, relational databases, flat file databases, object-relational databases, object-oriented databases, and any other structured collection of records or data that is stored in a computer system. Examples of RDBMS's include, for example, and without limitation, Oracle® Database (Oracle is a registered trademark of Oracle Corporation, Redwood Shores, Calif), MySQL, IBM® DB2 (IBM is a registered trademark of International Business Machines Corporation, Armonk, N.Y.), Microsoft® SQL Server (Microsoft is a registered trademark of Microsoft Corporation, Redmond, Wash.), Sybase® (Sybase is a registered trademark of Sybase, Dublin, Calif.), and PostgreSQL® (PostgreSQL is a registered trademark of PostgreSQL Community Association of Canada, Toronto, Canada). However, any database may be used that enables the systems and methods to operate as described herein.

[0188] Certain embodiments are described herein as including logic or a number of routines, subroutines, applications, or instructions. These may constitute either software (e.g., code embodied on a machine-readable medium or in a transmission signal) or hardware. In hardware, the routines, etc., are tangible units capable of performing certain operations and may be configured or arranged in a certain manner. In example embodiments, one or more computer systems (e.g., a standalone, client or server computer system) or one or more hardware modules of a computer system (e.g., a processor or a group of processors) may be configured by software (e.g., an application or application portion) as computer hardware that operates to perform certain operations as described herein.

[0189] In various embodiments, computer hardware, such as a processor, may be implemented as special purpose or as general purpose. For example, the processor may comprise dedicated circuitry or logic that is permanently configured, such as an application-specific integrated circuit (ASIC), or indefinitely configured, such as a field-programmable gate array (FPGA), to perform certain operations. The processor may also comprise programmable logic or circuitry (e.g., as encompassed within a general-purpose processor or other programmable processor) that is temporarily configured by software to perform certain operations. It will be appreciated that the decision to implement the processor as special purpose, in dedicated and permanently configured circuitry, or as general purpose (e.g., configured by software) may be driven by cost and time considerations.

[0190] Accordingly, the term “processor” or equivalents should be understood to encompass a tangible entity, be that an entity that is physically constructed, permanently configured (e.g., hardwired), or temporarily configured (e.g., programmed) to operate in a certain manner or to perform certain operations described herein. Considering embodiments in which the processor is temporarily configured (e.g., programmed), each of the processors need not be configured or instantiated at any one instance in time. For example, where the processor includes a general-purpose processor configured using software, the general-purpose processor may be configured as respective different processors at different times. Software may accordingly configure the processor to constitute a particular hardware configuration at one instance of time and to constitute a different hardware configuration at a different instance of time.

[0191] Computer hardware components, such as transceiver elements, memory elements, processors, and the like, may provide information to, and receive information from, other computer hardware components. Accordingly, the described computer hardware components may be regarded as being communicatively coupled. Where multiple of such computer hardware components exist contemporaneously, communications may be achieved through signal transmission (e.g., over appropriate circuits and buses) that connect the computer hardware

components. In embodiments in which multiple computer hardware components are configured or instantiated at different times, communications between such computer hardware components may be achieved, for example, through the storage and retrieval of information in memory structures to which the multiple computer hardware components have access. For example, one computer hardware component may perform an operation and store the output of that operation in a memory device to which it is communicatively coupled. A further computer hardware component may then, at a later time, access the memory device to retrieve and process the stored output. Computer hardware components may also initiate communications with input or output devices, and may operate on a resource (e.g., a collection of information).

[0192] The various operations of example methods described herein may be performed, at least partially, by one or more processors that are temporarily configured (e.g., by software) or permanently configured to perform the relevant operations. Whether temporarily or permanently configured, such processors may constitute processor-implemented modules that operate to perform one or more operations or functions. The modules referred to herein may, in some example embodiments, comprise processor-implemented modules.

[0193] Similarly, the methods or routines described herein may be at least partially processor implemented. For example, at least some of the operations of a method may be performed by one or more processors or processor-implemented hardware modules. The performance of certain of the operations may be distributed among the one or more processors, not only residing within a single machine, but deployed across a number of machines. In some example embodiments, the processors may be located in a single location (e.g., within a home environment, an office environment or as a server farm), while in other embodiments the processors may be distributed across a number of locations.

[0194] Unless specifically stated otherwise, discussions herein using words such as “processing,” “computing,” “calculating,” “determining,” “presenting,” “displaying,” or the like may refer to actions or processes of a machine (e.g., a computer with a processor and other computer hardware components) that manipulates or transforms data represented as physical (e.g., electronic, magnetic, or optical) quantities within one or more memories (e.g., volatile memory, non-volatile memory, or a combination thereof), registers, or other machine components that receive, store, transmit, or display information.

[0195] As used herein, the terms “comprises,” “comprising,” “includes,” “including,” “has,” “having” or any other variation thereof, are intended to cover a non-exclusive inclusion. For example, a process, method, article, or apparatus that comprises a list of elements is not necessarily limited to only those elements but may include other elements not expressly listed or inherent to such process, method, article, or apparatus.

[0196] Although the disclosure has been described with reference to the embodiments illustrated in the attached figures, it is noted that equivalents may be employed, and substitutions made herein, without departing from the scope of the disclosure as recited in the claims.

[0197] Having thus described various embodiments of the disclosure, what is claimed as new and desired to be protected by Letters Patent includes the following:

Claims

1. A computer-implemented method for enabling a secure remote commerce (SRC) transaction for charging an electric vehicle (EV) at a charging point (CP), the method comprising: establishing, by a user system of the EV, a mutually authenticated and encrypted communication session with the CP via a transport layer security (TLS) protocol; receiving, by the user system, merchant information from the CP, the merchant information including a unique identifier (ID) of the CP and a charge point operator certificate issuer identifier; retrieving, by the user system, payment credentials from a digital wallet stored in the EV, the payment credentials comprising a digital card

ID assigned to the EV by an SRC system; generating, by the user system, a cryptographically signed payload comprising the digital card ID, the merchant information, and a timestamp, the signed payload being signed by the user system using a private key associated with the EV; transmitting, by the user system, the signed payload to the SRC system via an application programming interface (API) over a secure communication channel; receiving, by the user system, an authorization response from the CP, the authorization response indicating that the payment transaction is approved; and in response to receiving an authorization approval, transmitting, by the user system, a command to the CP to initiate charging of the EV.

2. The method of claim 1, further comprising: binding the digital card ID to a public key associated with the EV via a binding API before initiating the transaction.

3. The method of claim 1, the operation of generating the cryptographically signed payload comprises applying a digital signature using an asymmetric key pair.

4. The method of claim 1, further comprising: monitoring for a payment authorization response via the CP, and upon receiving a positive payment authorization response, permitting the CP to deliver charging power to the EV.

5. The method of claim 1, the operation of establishing the mutually authenticated and encrypted communication session comprises using mutual TLS (mTLS) authentication, including transmitting, by the user system, a user system digital certificate to the CP and receiving, by the user system, a CP digital certificate.

6. The method of claim 1, the operation of receiving the merchant information from the CP includes receiving the merchant information as part of an ISO 15118-compliant communication session.

7. The method of claim 1, the digital card ID comprising a tokenized representation of a primary account number (PAN).

8. The method of claim 1, further comprising: transmitting, by the user system, an authorization request to the CP, requesting commencement of charging of the EV.

9. The method of claim 1, further comprising: querying the CP, by the user system, for available network services to facilitate payment processing in the absence of a direct network connection by the EV.

10. The method of claim 1, further comprising: receiving, by the user system, the digital card ID through a secure enrollment process with the SRC system, the enrollment process including: establishing, by the user system, a secure API connection with the SRC system over a secure communication channel, transmitting, by the user system, an enrollment request to the SRC system, the enrollment request including payment card information and user identification data; receiving, by the user system, an authentication challenge from an issuer associated with the payment card information; transmitting, by the user system, an authentication input verifying the user's identity in response to the authentication challenge; and receiving, by the user system, the digital card ID from the SRC system.

11. A computing system for enabling a secure remote commerce (SRC) transaction for charging an electric vehicle (EV) at a charging point (CP), the system comprising: one or more processors; and a memory having computer-executable instructions stored thereon, that when executed by the one or more processors, cause the one or more processors to perform operations of: establishing a mutually authenticated and encrypted communication session with the CP via a transport layer security (TLS) protocol; receiving merchant information from the CP, the merchant information including a unique identifier (ID) of the CP and a charge point operator certificate issuer identifier; retrieving payment credentials from a digital wallet stored in the EV, the payment credentials comprising a digital card ID assigned to the EV by an SRC system; generating a cryptographically signed payload comprising the digital card ID, the merchant information, and a timestamp, the signed payload being signed by using a private key associated with the EV; transmitting the signed payload to the SRC system via an application programming interface (API) over a secure

communication channel; receiving an authorization response from the CP, the authorization response indicating that the payment transaction is approved; and in response to receiving an authorization approval, transmitting a command to the CP to initiate charging of the EV.

12. The system of claim 11, the computer-executable instructions further cause the one or more processors to perform an operation of binding the digital card ID to a public key associated with the EV via a binding API before initiating the transaction.

13. The system of claim 11, the operation of generating the cryptographically signed payload comprises applying a digital signature using an asymmetric key pair.

14. The system of claim 11, further comprising: the computer-executable instructions further cause the one or more processors to perform operations of: monitoring for a payment authorization response via the CP, and upon receiving a positive payment authorization response, permitting the CP to deliver charging power to the EV.

15. The system of claim 11, the operation of establishing the mutually authenticated and encrypted communication session comprises using mutual TLS (mTLS) authentication, including transmitting a user system digital certificate to the CP and receiving, by the user system, a CP digital certificate.

16. The system of claim 11, the operation of receiving the merchant information from the CP includes receiving the merchant information as part of an ISO 15118-compliant communication session.

17. The system of claim 11, the digital card ID comprising a tokenized representation of a primary account number (PAN).

18. The system of claim 11, the computer-executable instructions further cause the one or more processors to perform an operation of transmitting an authorization request to the CP, requesting commencement of charging of the EV.

19. The system of claim 11, the computer-executable instructions further cause the one or more processors to perform an operation of querying the CP for available network services to facilitate payment processing in the absence of a direct network connection by the EV.

20. The system of claim 11, the computer-executable instructions further cause the one or more processors to perform an operation of receiving the digital card ID through a secure enrollment process with the SRC system, the enrollment process including: establishing a secure API connection with the SRC system over a secure communication channel, transmitting an enrollment request to the SRC system, the enrollment request including payment card information and user identification data; receiving an authentication challenge from an issuer associated with the payment card information; transmitting an authentication input verifying the user's identity in response to the authentication challenge; and receiving the digital card ID from the SRC system.
